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THE LINKS BETWEEN GLOBAL VALUE CHAINS AND GLOBAL INNOVATION NETWORKS

AN EXPLORATION

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FOREWORD

Economic globalisation has dramatically increased during the past decades and as a result, economic activities increasingly stretch out across OECD and emerging economies. Goods, services, capital, people, technology and knowledge are easily transferred across national borders. Following the international fragmentation of production, goods and services are nowadays produced and heavily traded in international production networks or Global Value Chains (GVCs). More recently, innovation activities have also become increasingly internationalised thereby giving rise to global innovation networks (GINs). The networks typically consist of own R&D facilities abroad as well as collaborative arrangements with external partners and suppliers.

The nexus between these two types of networks in the global economy has not really been explored, although strong interdependencies between GVCs and GINs are likely. For example, do the two types of networks show a similar geographic and industry concentration; more specifically, are hubs and nodes of GVCs the same as in GINs? Do countries that trade with each other within GVCs show also a propensity to cooperate in GINs? Are countries that are involved in trade and production networks more likely to innovate with each other? Does the co-operation in innovation and GINs result in better economic performance in GVCs?

This paper takes a first attempt to analyse the linkages between both types of networks and identify a number possible government implications. The motivation for this analysis is that concerns are raised in policy discussions that countries are not able to capture the value of their innovative activities.

ABSTRACT

Economic globalisation has given rise to two types of networks that stretch out across OECD and emerging economies. At the one side, Global Value Chains (GVCs) can be thought of as the "material" transfers of goods and services (final as well as intermediate) across borders. At the other side, Global Innovation Networks (GINs) refer to the transfers of intangibles and immaterial assets between countries. Concerns are increasingly raised in policy discussions that countries are not able to capture the value of their innovative activities, hence the clear need to better understand the interdependencies between these two types of networks.

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1. Introduction: the globalisation of economic activities

During the past decades, economic globalisation has dramatically progressed with economic activities increasingly stretching out across OECD and emerging economies. Goods, services, capital, people, technology and knowledge are increasingly moving across national borders. Multinational companies (MNEs) have been a driving force of growing globalisation, but more purely “domestic’ actors like SMEs, universities, public research organisations, etc. have also expanded the geographical reach of their activities. Globalisation increasingly affects how a growing number of economic actors operate, compete and innovate, both at home and abroad.

The international trade of goods and more recently also of services has been the most prominent channel of economic globalisation. For decades, final products have been produced in one country and were then exported to other countries according to the principle of comparative advantage. In more recent decades international trade has grown further in importance as individual stages in the production process are increasingly dispersed across different countries. This has resulted in the growing international trade of intermediate goods within global value chains (GVCs), in addition to trade of final and capital goods. Trade and investment are closely linked in GVCs as MNEs have been the first to internationalise their production networks with some affiliates having a world or regional mandate for the production of a specific product/service. OECD (2013a) describes in detail the emergence of these international production networks and gives an overview of their implications on different policy domains.

More recently, innovation activities have also become increasingly internationalised giving rise to global innovation networks (GINs); nevertheless, it should be noted that the majority of R&D investments is still concentrated in the home country near the headquarters (OECD, 2008a; Belderbos et al., 2016). Following the offshoring of production activities, companies have also started to offshore R&D activities with MNEs increasingly setting up R&D activities abroad. To support their international production and distribution activities; these R&D investments abroad were in the beginning mainly designed to adapt products and process to local market demands. But R&D investments have recently also been motivated to tap into foreign knowledge, technology and human capital, and as such leverage the innovation activities which remain concentrated in R&D headquarters at home.

In addition, a trend of “open” innovation has been observed with companies opening their innovation process to external partners and outsourcing innovation activities, often abroad (OECD, 2008b). Owing to more intense and global competition and technological progress, product life cycles have been drastically shortened, forcing companies to innovate more quickly and develop products and services more efficiently. Moreover, the growing integration of different technologies has made innovation more costly and riskier. The greater the need for interdisciplinary cross-border and cross-sector research, the less a single company has the capability to innovate successfully. Companies increasingly look for partners with complementary expertise to obtain access to different technologies and knowledge quickly.

These GINS include own R&D facilities abroad as well as collaborative arrangements with external partners and suppliers, in which firms depend in various ways on the expertise of the different partners. Companies link into these GINs with people, institutions (universities, government agencies, etc.) and other companies in their own or different countries to solve problems, source knowledge and generate ideas. GINs influence national and regional innovation systems. MNEs' ecosystems or networks of innovation often represent "nodes" linking regional/national systems of innovation across borders and therefore various S&T actors in different countries.

Until now, and maybe surprisingly, not much analysis has been undertaken to explore the nexus between these two types of networks[1], one reason being the lack of accurate and complete data (see below). In order to identify the likely interdependencies between GVCs and GINs – the first one representing more "material" transfer across borders, while the second refers more to the transfer of intangibles and immaterial assets, this paper takes a first attempt to analyse the nexus between both types of networks and identify a number possible government implications:

- First, how do GVCs and GINs map onto each other? Do the two types of networks show a similar geographic and industry concentration; more specifically, are hubs and nodes of GVCs the same as in GINs?

- Second, do countries that trade with each other within GVCs show also a propensity to cooperate in GINs? Are countries that are involved in trade and production networks more likely to innovate with each other? If so, non-technological policies focusing on trade promotion may also help to encourage international co-operation in innovation;

- Third, does the co-operation in innovation and GINs result in better economic performance in GVCs? It has been shown that co-operation in innovation has positive effects on technological outcomes and in some cases enhances economic performance (e.g. productivity), but does it also have an impact on countries' performance in GVCs, in particular on the upgrading trajectories within GVCs?

The motivation for casting light on these questions relates to concerns raised in policy discussions that countries are not able to capture the value of their innovative activities.

In providing a (partial) answer to these questions, this paper complements other CIIE work focusing on the link between Knowledge Based Capital (KBC) and GVCs. That work aims to identify the interdependencies between KBCs and GVCs in a broad sense, covering different forms of KBC (R&D, patents, organisational capital, etc.). By focusing on GINs, this paper focuses specifically on the importance of international co-operation in innovation.

2. Mapping of GVCs and GINs

2.1. Empirical measures of GVCs and GINs

OECD work in recent years, and particularly the construction of the Trade in Value Added (TiVA) database[2], has provided the evidence base necessary for the analysis of GVCs. Before this, the emergence of international production networks was mainly described through case studies and company/industry surveys. The new TiVA data, however, document the participation and upgrading of countries in GVCs across industries and over time. Among other contributions, the work has shown that GVCs have become more important and complex, with more and more countries participating in these international production networks (OECD, 2013). De Backer and Miroudot (2013) map GVCs across countries, industries and time and show that the length of GVCs has significantly increased over time with different roles for different countries. In addition, they show how the geography of GVCs has changed over the years with emerging economies taking up a growing role in a number of industries.

The evidence on GINs is not as widely available, and furthermore GINs have largely been discussed in terms of case studies typically in high-technology sectors. Recently also more aggregate indicators based on R&D data, patent data, innovation surveys have increasingly been used to describe the growing importance of open innovation [3] – for an overview see OECD (2008b).

- Data on R&D investments are internationally comparable, but only provide rather indirect evidence on open innovation. For example, specific information on public-private funding of R&D reveals some of the interactions and collaboration between government and the business sector. Government-financed business R&D is typically more related to direct government funding, than to actual collaboration between government and business. In contrast, business funding in the higher education and government sectors (*e.g.* research centres) often indicates close collaboration between public and private entities.

- Innovation surveys are increasingly used in OECD and in many non-member countries to better understand the role of innovation and the characteristics of innovative companies. The latest surveys include information on innovation linkages, including collaboration on innovation. Collaboration, defined as the "active participation in joint innovation projects with other organisations" may involve the joint development of new products, processes or other innovations with customers and suppliers, as well as horizontal work with other enterprises or public research bodies. Therefore, more direct evidence on open innovation and specifically on the sourcing of innovation can be derived from innovation surveys.

- Patent data are considered a unique, broadly available and reliable source of statistical material and are increasingly used to study different aspects of the innovation process since they represent an output of knowledge creation and innovation (Griliches, 1990; OECD, 2009).[4] Patent data provide detailed information such as the nature of the invention as well as the actors involved in the innovative process. Patent documents report the inventor(s), the applicant(s) – i.e. the owner of the patent at the time of application – along with their addresses and countries of residence. Furthermore, the time series nature of patent data allows for an analysis of innovative activities over time. The main disadvantage of patent statistics is that they fail to capture all innovative activity, as not all innovations are patented and not all patents lead to innovations. Typically, patent information on different inventors, different co-assignees or owners, differences between inventors and assignees are used to analyse open innovation and identify GINs.

Box 1. Some tentative conclusions about open innovation and GINs

Indicators on open innovation based on large-scale data suggest that companies increasingly innovate together with external and international partners. The industry distribution shows that collaboration on innovation is significant in manufacturing as well as in services, although certain industries (chemicals, pharmaceuticals, ICT, including software) typically have higher levels of open innovation. While co-operation in innovation is on the rise, the data show clearly that larger firms innovate more openly than SMEs. These results suggest that limited resources may prevent SMEs to deploy open innovation practices more broadly and on an international scale. Large companies are much more active in public research although there is more cross-country variation for large firms than for SMEs.

Companies collaborate on innovation with suppliers and customers more than with universities and government research institutions, at least in terms of numbers of collaborations. This may be because public research focuses more on upstream research and exploration activities which may be a small part of overall innovation.

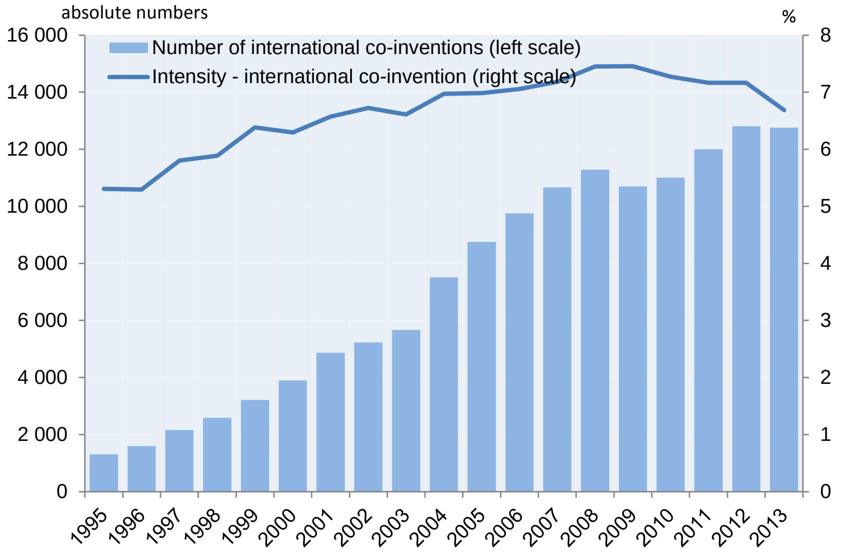
The empirical data also show that despite globalisation, geographic proximity still matters in open innovation. Companies were found to collaborate more with geographically close external partners, although it should be noted that the data measure the number of interactions and not the intensity and quality of collaboration. Additional evidence suggests that proximity may matter somewhat less than good connectivity with external partners.

Source: OECD (2008b); see also OECD Science, Technology and Industry Scoreboard 2013 and 2015.

This paper uses patent data and in particular information on international co-invention to identify and analyse GINs. It is clear from the start however that this information – just like all indicators described above - does not allow for the identification of all types of GINs across countries and industries. International co-invention refers to patents with multiple inventors who reside in different countries. The patent data used in this paper refer to patent applications filed under the Patent Co-operation Treaty (PCT) at the EPO, in order to guarantee the international comparability across countries[5]. Addresses of the inventors are used to assign the patent applications to countries while PCT patents are assigned to industries based on the applicant information[6] using the Harmonized Applicants' Names (HAN) database. This database provides a depository of the names connected to the applicant (and not the inventor) and is expanded upon with business register data.

After dropping equivalent PCT patents – i.e. through the identification of patents with the same inventors and same priority application date – almost 2 million PCT patents are obtained for the period between 1995 and 2013. Amongst these, 137,758 international co-inventions were identified based on the addresses of inventors. Figure 1 shows the level of international co-operation in innovation across national borders between 1995 and 2013. As can be seen clearly, the trend has generally been increasing, at the time of the economic crisis in 2008-2009 international co-invention decreased globally before recovering (see also Guellec and Van Pottelsberghe, 2001 for a discussion of earlier years). Moreover, the co-invention intensity (i.e., the number of co-inventions over the total number of PCT patents) has actually decreased since the economic crisis after a gradual rise since 1995.

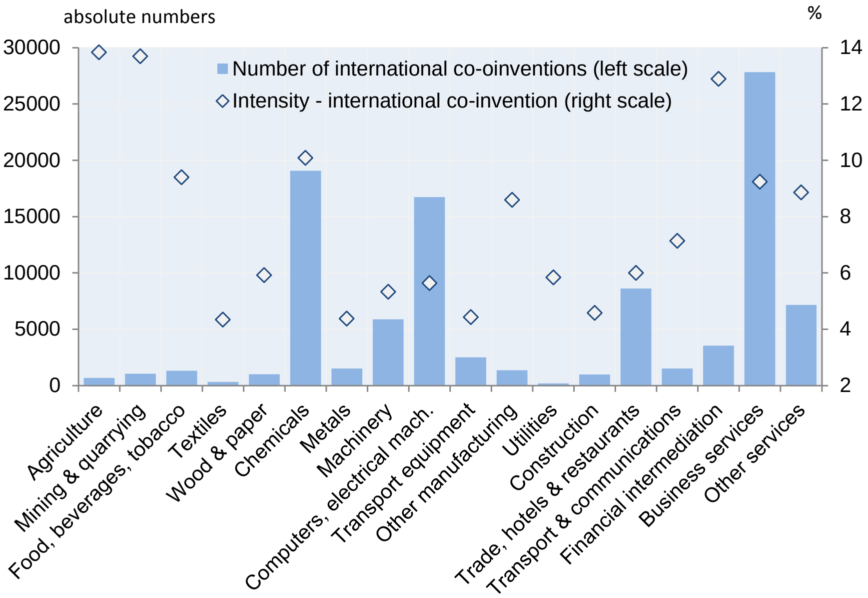
Figure 1. International co-invention, absolute numbers and intensity, 1995-2013



Note: Intensity is measured as the number of international co-inventions in percentage of total number of PCT applications.
Source: calculations based on OECD patent database.

Important differences in co-invention exist across industries reflecting the fact that some industries show a higher propensity to patent and a higher proportion of international co-invention. A large number of international co-inventions are observed in chemicals, electronics and business services (Figure 2); other industries with a significant number of international co-inventions are machinery, whole/retail trade, hotels and restaurants and other services. Intensities (international co-inventions as percentage of PCT patent) show a somewhat different pattern with high intensities in a number of industries with a relatively small number of PCT patents: agriculture, mining, food and financial intermediation. For the other industries, the intensity varies in general between 4 and 10%.

Figure 2. International co-invention across industries, absolute numbers and intensity, 1995-2013

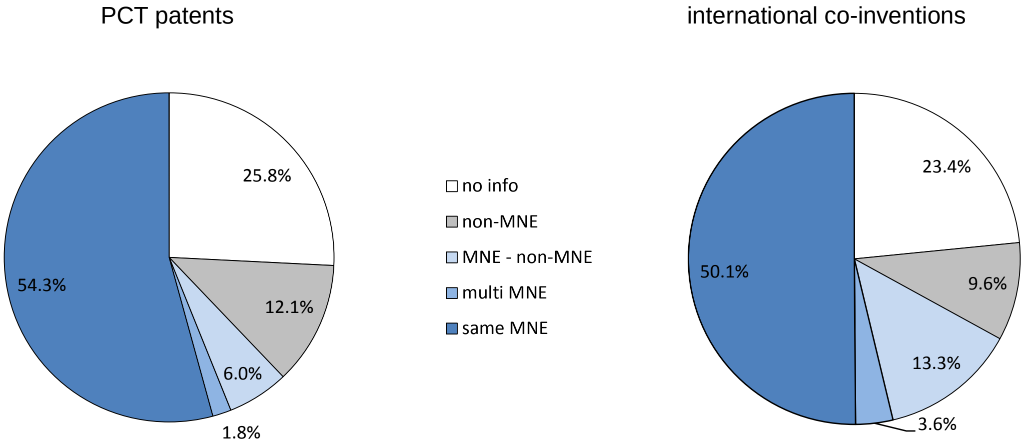


Note: Intensity is measured as the number of international co-inventions in percentage of total number of PCT applications.
Source: calculations based on OECD patent and HAN databases.

Given the central role of MNEs in cross-border activities in general and GINs in particular, it can be expected that co-invention patterns across countries are partially driven by the activities of MNEs. Based on the applicants' information in terms of ownership structure[7], PCT patents and co-inventions are allocated to MNEs and non-MNEs.[8] The results clearly show the importance of MNEs in both patenting in general, as well as international co-invention (Figure 3): more than 60% of all PCT patent applications are found to be related to MNE activities (i.e. where an MNE –headquarter or affiliate – was the only or one of the applicants of the patent).

For co-inventions, the importance of MNEs is even more marked with two-thirds of the co-inventions that can be directly linked to MNE activities. In particular, more than 50% concern co-inventions between inventors in different countries but with the same MNE as applicant (i.e. headquarters and/or affiliates). Thirteen percent concern co-inventions between at least one MNE and at least one non-MNE, while 5% concern co-invention between different groups of MNEs.[9]

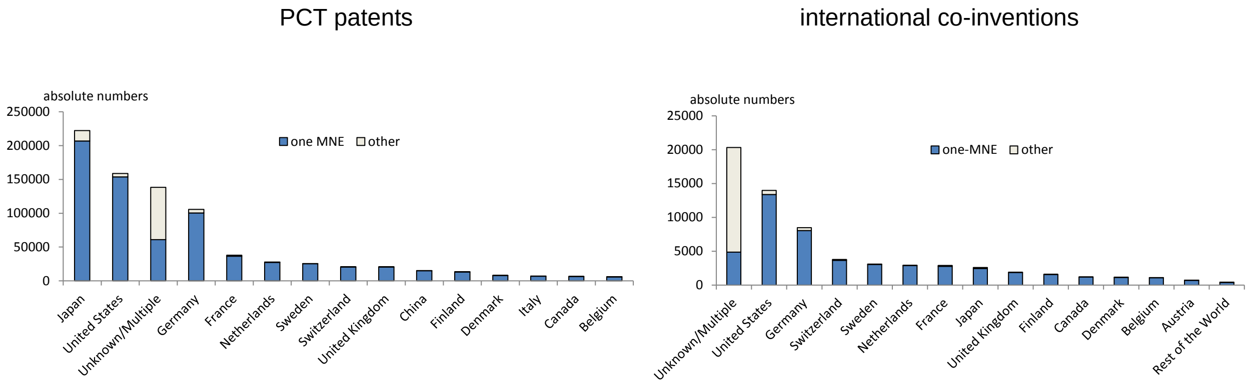
Figure 3. The importance of MNEs in patenting and international co-invention, 1995-2013



Note: The percentages present the share of each group in the total number of PCT applications and international co-inventions
Source: calculations based on OECD patent, HAN and ORBIS databases.

Figure 4 presents the number of PCT patents and co-inventions according to the nationality of the MNEs (Top 15 countries, based on the global ultimate owner).[10] MNEs with headquarters in Japan, the United States and Germany are observed to have most of PCT patent applications between 1995 and 2013 across the globe.[11] But also smaller countries like Sweden, Switzerland and the Netherlands – i.e. countries with a large presence of domestic MNEs – occupy an important position. In terms of international co-inventions, MNEs from the United States, Germany and Switzerland make up the top-3 largest applicants; the large majority of these applications are held within the same MNE group.

Figure 4. Top 15 MNE home countries in patenting and international co-invention, 1995-2013



Source: calculations based on OECD patent, HAN and ORBIS databases.

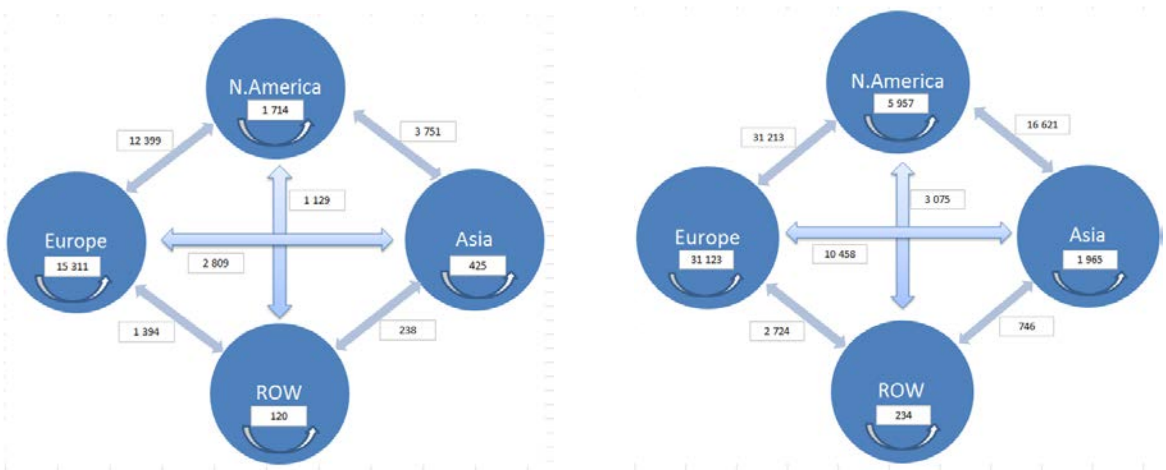
2.2. The geography of co-invention and GINs

The geography of co-invention has significantly changed over the past decades with GINs becoming denser and increasingly stretching out over a growing number of countries, including emerging economies. A first look at international co-invention across different regions shows the importance of North America and Europe for international co-operation in innovation: together they were responsible for more than 80% of all international co-inventions between 1995 and 2013 (Figure

5). This is due to their innovation links with other regions but also strong intra-regional co-invention (particularly for Europe).[12] Between the two sub-periods (1995-2004 and 2005-2013), both regions witnessed a strong increase in the number of international co-inventions, again both intra- and extra-regionally. In addition, the growing importance of Asia is clearly observed between the two sub-periods as Asian partners increasingly collaborate in innovation with North American and to a lesser extent with European partners. The intra-regional co-inventions in Asia (i.e. international collaborations among Asian countries) also increased over time, but less strongly than the extra-regional links of Asia. This growing importance of Asia in co-invention patterns reflects the importance of Asian markets for companies in Europe and North America as well as the growing investments in Science, Technology and Innovation (STI) in these countries.

Figure 5. International co-invention by region (absolute numbers), 1995-2004 and 2005-2013

1995-2004 2005-2013

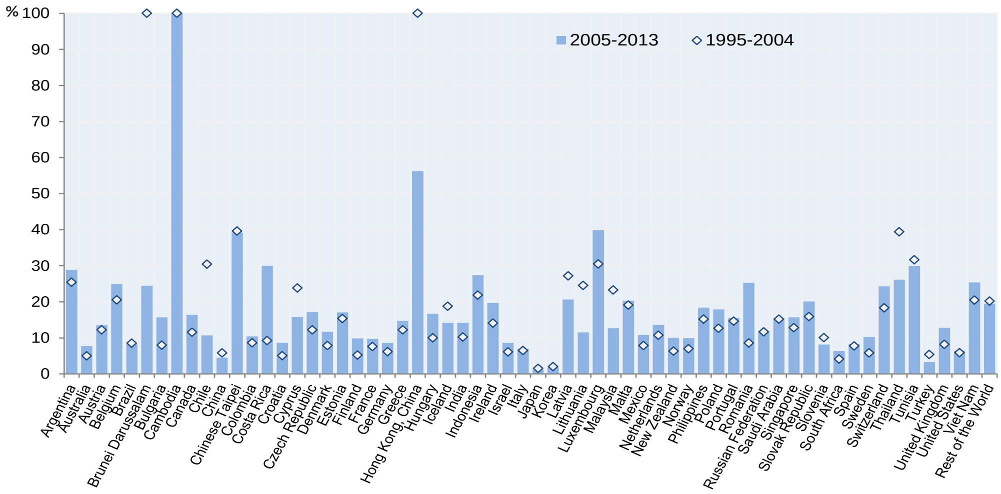


Source: calculations based on OECD patent database.

Moving to the individual country level, a country's degree (or intensity) of international co-invention can be measured as the number of patents invented by a country with at least one foreign inventor in the total number of patents invented domestically (see also Hascic et al., 2012). As suggested by Dernis and Khan (2004), co-invention for individual countries is calculated by fractional counts depending on the number of inventors and their countries of residence in order to prevent double counting. To illustrate this, imagine a hypothetical scenario where a patent consists of two inventors in two different countries. In this case, each country will be allocated a share of 0.50. If however there are three inventors all from different countries, each country would allocate a share of 0.33 and so on. In addition, the more inventors from a specific country, the greater their share for that country becomes. For example, a patent with 4 inventors, two from the United States, one from Germany and one from the United Kingdom, the countries shares are 0.5, 0.25 and 0.25, respectively.

Figure 6 presents the intensity of the different countries in the sample: in line with the growing trend of international co-invention during the past decades, most of the countries have increased their co-invention share between 1995 and 2013. Japan and Korea show very low international co-invention shares indicating the domestic focus of their innovation process (at least in patenting). Often smaller countries show relatively higher co-invention shares. Also remarkable is the strong co-invention share of a number of emerging economies, which is again most likely due to the growing attractiveness of these economies for other partners and the search of these economies for access to foreign knowledge.

Figure 6. International co-invention intensity by country, 1995-2013



Note: Intensity is measured as the number of international co-inventions in percentage of total number of PCT applications.
Source: calculations based on OECD patent database.

Analysing the above country co-invention measures by partner allows for a more detailed assessment of changes in the geography of GINs; in addition similar trade intensity by partner have been calculated to start to reveal the links between GINs and GVCs.

The following co-invention and trade intensity measures therefore attempt to illustrate the innovation and trade networks between countries within industries overtime[13]:

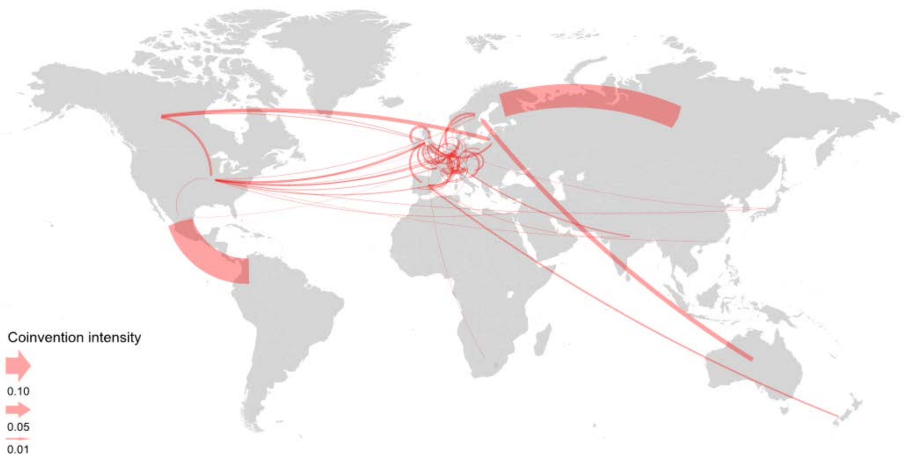
$$CoinvIntensity = \left(\frac{Coinvention_{ijk,t} + Coinvention_{jik,t}}{\sum Invention_{ikt} + \sum Invention_{jkt}} \right) \quad (1)$$

$$TradeIntensity = \left(\frac{Exports_{ijk,t} + Exports_{jik,t}}{\sum Output_{ikt} + \sum Output_{jkt}} \right) \quad (2)$$

Figure 7 presents maps for three separate industries (one manufacturing industry with a low number of international co-inventions – textiles; one manufacturing industry with a high number of international co-inventions – electric and optical equipment and one services industry – real estate, renting and business services).[14] The graphs for the other industries can be found in Annex 1.[15]

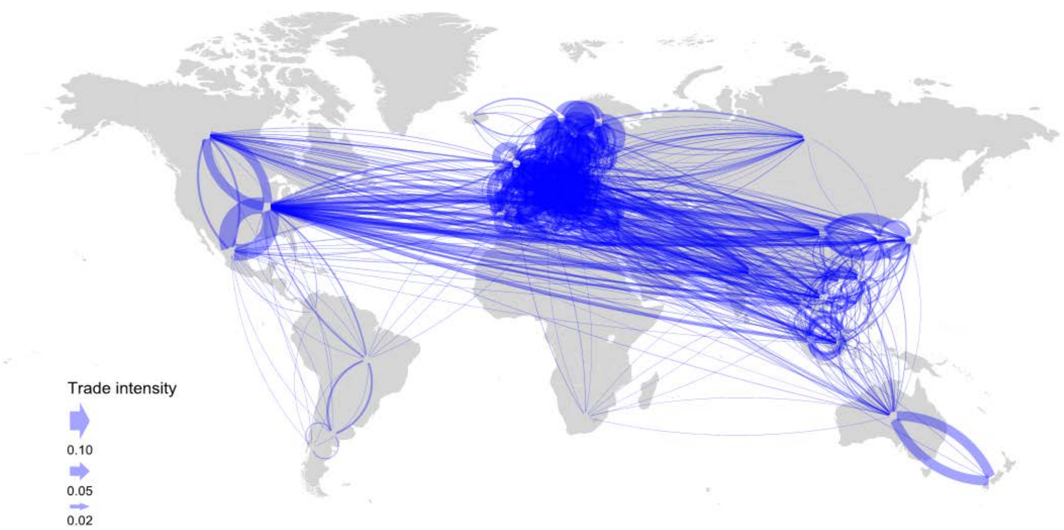
Figure 7. Co-invention and trade intensity, by country, partner and industry 1995-2013

Textiles, textile products, leather and footwear (ISIC 17 to 19), 2000-05



International co-invention intensity[1]

Trade intensity[2]



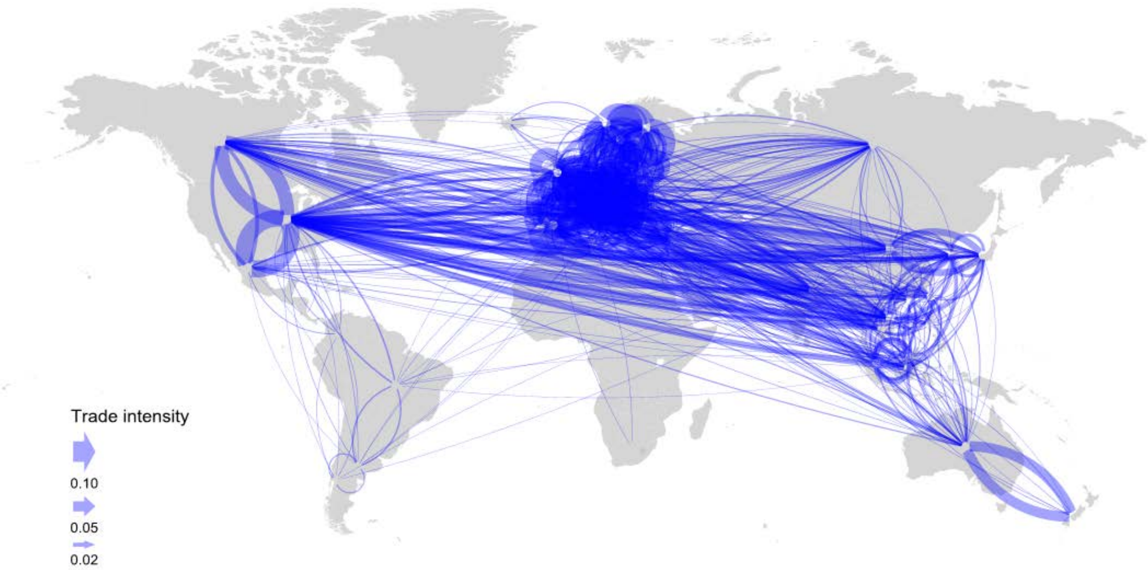
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Textiles, textile products, leather and footwear (ISIC 17 to 19), 2008-11

International co-invention intensity[1]

Trade intensity[2]



1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Electrical and optical equipment (ISIC 30 to 33), 2000-05

International co-invention intensity[1]

Trade intensity[2]

1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Electrical and optical equipment (ISIC 30 to 33), 2008-11

International co-invention intensity[1]

Trade intensity[2]

1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Real estate, renting and business activities (ISIC 70 to 74), 2000-05

International co-invention intensity[1]

Trade intensity[2]

1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Real estate, renting and business activities (ISIC 70 to 74), 2008-11

International co-invention intensity[1]

Trade intensity[2]

1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Source: calculations based on OECD patent database

A number of common observations emerge from the figures:

a) GINs have become more important over time, as reflected in the growing links between countries. In addition, a clear broadening of GINs across countries has taken place: the number of countries engaging with multiple partners has substantially risen. In particular, South East Asia experienced a significant increase in participation within innovation networks. In contrast, countries in the African region and much of South America have shown little increase in engagement in patent collaboration, aside from Brazil and Chile.

b) The changes over time in international trade networks have overall been smaller, probably since these networks were already more established.

c) Trade networks in goods are denser than in services, with more trade flows in manufacturing than in services industries.[16] The difference between manufacturing and services seems to be less marked in innovation networks, with for example in the services industry of ‘renting, real estate and business services’ a large and growing number of international co-inventions.

d) When comparing both types of networks, trade networks appear to be more intensive than GINs: trading relationships show more and ‘thicker’ links between countries than co-invention networks. GINs have become more intensive in the second sub-period but not up to the same level as GVCs.

e) Overall, GINs show a more concentrated pattern with a small number of partners cooperating in innovation. GVCs are more dispersed and countries trade with more partners than they co-invent. Both types of networks show a strong regional character suggesting that distance (still) seems to matter in GVCs as well as GINs.

f) A significant geographic overlap exists between GINs and GVCs, taking into account the differences discussed above. This reflects the strong concentration of international co-inventions and trade across the large regions, with important hubs in the United States, Europe (around Germany and France) and East Asia (around Japan, the People’s Republic of China and Korea). The majority of intra- and extra-regional links are also centred around these hubs

3. From GVCs to GINs: does trade promote co-invention?

Following the mapping of GVCs and GINs, a more formal analysis is proposed to assess the potential interdependencies between both types of networks. In a first analysis, the effect of GVCs on GINs is analysed: does co-operation (i.e. trading) in GVCs increase the propensity and extent of co-invention between countries. More specifically: do countries that trade with each other in GVCs show also higher co-invention rates with each other in GINs; do countries that start to trade more intensively subsequently exhibit higher co-invention rates.? One reason for this link is that through establishing trade relationships companies/countries may acquire contacts and get to know potential partners for co-operating in innovation. In addition, working together in one type of transaction makes partners more familiar with each other and may induce co-operation in other transactions, particularly in innovation where familiarity and trust can be expected to be more important than for example in trade.

In order to confirm this type of link between GVCs and GINs, a gravity-type[17] model is estimated explaining bilateral co-invention between countries at the industry level and relating this to a number of similar factors as in the usual gravity model on the one hand, as well as the trade relations between the countries on the other hand. This empirical strategy follows a number of papers that have analysed technological collaboration in a gravity context (Guellec and Van Pottelsberghe, 2001; Maggioni et al., 2007; Picci, 2010 and Montobbio and Sterzi, 2013). The dependent variable in the model is the number of bilateral co-inventions between each pair of countries per industry and year. Because of the large numbers of zeros – a considerable number of country pairs do not show international co-inventions, thus the gravity model has been estimated using the Poisson Pseudo Maximum Likelihood (PPML) estimator. The results are presented in Table 1.[18]

Patent stocks of country pairs are used as ‘attractors’ in the model, assuming that countries will collaborate more in innovation the larger their respective innovation activities in general.[19] Because of the limited availability of KBC variables - longitudinal at the industry level and for OECD as well as emerging economies, patent stock is used as a – albeit imperfect – measure of the innovation activities of countries.[20] The patent stock for each year has been calculated based on the number of new inventions and, in line with the existing literature, applying a 15% depreciation rate on the existing inventions.

Just like in the trade-gravity model, distance is assumed to moderate the probability and extent of co-invention, GINs are more difficult to develop between partners further away from each other. This paper uses a number of different ‘distance’ measures across models; in the first place, the traditional measure of physical distance expressed as the number of kilometres between country/partner pairs.[21] Because of the growing importance of communication technologies including broadband, satellite communications but also social networks, it has been argued that physical distance has however become less important. Instead, differences in time zones may still matter in today’s easy and low-cost communications world, as this would complicate and prevent real-time communication between partners. As such, variables are included on the hour difference in time zones, and a dummy variable for being in the same time zone.[22]

Also cultural and historical distances are taken into account as they have been found to affect the probability and intensity of collaboration. More specifically, indicators have been included to reflect whether or not countries speak the same language and if they share a national border or colonial history.

In addition, a measure of technological distance is included since previous research has demonstrated that ‘technologically’ close partners seem to collaborate more. Again, several measures have been proposed in the literature going back to the seminal work of Jaffe in 1989 with each measure possessing strengths and weaknesses (for an overview, see Stellner, 2014). Technological distance in this paper is calculated as the difference (in size) between the partners’ patent stocks; because of the non-directional dependent variable, the absolute value of this difference in included.

Last but not least, the trade relationship between partners is included as an additional measure of distance to analyse the interdependencies between GVCs and GINs. The hypothesis is that partners will collaborate more in innovation if they also collaborate in trade networks as they are more familiar with each other. ‘Trade’ distance is measured by the bilateral trade (exports + imports) from one country to the other country (at the industry level). As this variable also contains many zero values; a dummy variable is include in order to assess the effects of zero trade links on co-invention:

$$\begin{aligned} coinv_{ijst} &= \alpha_0 + \alpha_1 patstock_{ist} + \alpha_2 patstock_{jst} + \alpha_3 distance_{ij} + \alpha_4 tdiff_{ij} + \\ &\alpha_5 notdff_{ij} + \alpha_6 contiguity_{ij} + \alpha_7 language_{ij} + \alpha_8 colony_{ij} + \alpha_9 techdist_{ijst} + \\ &\alpha_{10} mne_{ist} + \alpha_{11} trade_{ijst} + \alpha_{12} notrade_{ijst} + \alpha_{13} \delta_{ijst} + \varepsilon \end{aligned} \tag{3}$$

with i = country i, j = partner country j, s = industry and t = year

Table 1. Co-invention gravity model: econometric results

	Base model a)	Base model b)	Base model c)	Model 1	Model 2	Model 3	Model 4
Patent Stock country	0.510***	0.537***	0.474***	0.474***	0.473***	0.469***	0.484***
	(0.01)	(0.01)	(0.03)	(0.03)	(0.03)	(0.03)	(0.06)
Patent Stock partner	0.452***	0.477***	0.274***	0.274***	0.274***	0.271***	0.373***
	(0.01)	(0.01)	(0.04)	(0.04)	(0.04)	(0.04)	(0.06)
Physical distance	-0.108***	-0.109***	-0.079***		-0.035*	-0.035*	-0.037
	(0.01)	(0.01)	(0.01)		(0.02)	(0.02)	(0.03)
Time difference				-0.067***	-0.041**	-0.041**	0.032
				(0.01)	(0.02)	(0.02)	(0.04)
Same time zone				0.161***	0.159***	0.159***	0.102*
				(0.03)	(0.03)	(0.03)	(0.05)
Contiguity (dummy)	0.327***	0.328***	0.381***	0.407***	0.378***	0.370***	0.371***
	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.05)
Language (dummy)	0.947***	0.965***	0.707***	0.669***	0.670***	0.665***	0.600***
	(0.03)	(0.03)	(0.02)	(0.03)	(0.03)	(0.03)	(0.04)
Colony (dummy)	0.053	0.024	-0.105***	-0.124***	-0.118***	-0.118***	-0.032
	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)
Technological distance						0.015***	0.001
						(0.01)	(0.01)
MNE Turnover							0.015**
							(0.01)
Bilateral trade	0.282***	0.264***	0.275***	0.271***	0.267***	0.269***	0.334***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
No bilateral trade	-2.967***	-2.893***	-3.879***	-3.843***	-3.850***	-3.847***	
	(0.71)	(0.71)	(0.75)	(0.75)	(0.75)	(0.75)	
Control Variables							
Year		✓	✓	✓	✓	✓	✓
Origin/Destination fe's			✓	✓	✓	✓	✓
R-squared	0.650	0.664	0.889	0.891	0.891	0.891	0.926
Observations	199,534	199,534	174,132	174,132	174,132	173,918	35,670

Note: No bilateral trade is a dummy which equals to one when there is zero trade (imports + exports) from a country to a partner in a particular industry and zero otherwise. Level of significance * p<0.10, ** p<0.05, *** p<0.01

The base co-invention model shows similar results as in the typical trade-gravity equations[23]: Co-invention is proportional to the innovative activities of the partners and inversely proportional to the distance between the countries. The size of the innovative activities reflected in the patent stock of the countries is observed to be an important determinant of international knowledge links (see also Montobbio and Sterzi, 2013). Also historical and cultural ties play a role: speaking the same language and sharing a common border is positively related to global co-invention. The effect of colonial history between countries on co-invention indicates a negative effect which may be due to the different levels of economic development between country pairs.

Consistent with much of the literature on distance and knowledge diffusion, there appears to be less international collaboration the greater the distance between countries; both the physical distance in kilometres (base model and model 2) as well as the difference in time zones (models 1 and 2) negatively relates to the propensity and extent of co-invention. On the basis of the size of the estimated coefficients, the results suggest that time difference is somewhat more important than geographic distance. Furthermore, operating in the same zone strongly and positively correlates to co-invention between partners. In sum, both face-to-face interaction costs (as proxied by geographic distance) and virtual interaction costs (as proxied by time distance) seem to matter for GINs (Montobbio and Sterzi, 2013).

In contrast to previous research on co-invention, the results for technological proximity/distance (model 3) indicate that co-invention is more likely the larger the technological gap between the partners. Technology sourcing may be one explanation for this result, with less innovative countries purposefully setting up technology links with stronger partners in order to access their knowledge and learn from these more knowledgeable partners. More research is needed in order to confirm this motivation as the chosen indicator – difference in size of patent stocks – may be an imperfect measure of technological distance/proximity.

The results demonstrate, unlike Montobbio and Sterzi (2013), a strong and positive correlation between bilateral trade and co-invention suggesting the interdependence between GVCs and GINs at the country level. In addition, having no trade relationship with a partner country strongly decreases the propensity and extent of co-invention with that partner. The results thus demonstrate that being in a trade network – be it through exporting and/or importing - provides countries with information on potential partners and/or makes them more familiar with these partners; this positively impacts the likelihood of cooperating in innovation.

The positive effect of trade on co-invention is also prominent in all the different models when accounting for the importance of MNEs across industries and countries. As MNEs have been observed to play an important role for the international co-invention links (see above) as well as for the trade relationship and specialisation of (some) countries (see. Caves, 1996; Sleuwagen and De Backer, 2001), the link between GVCs and GINs could be solely driven by MNE activities. This does not seem to be the case given that the export variable still exhibits a significant effect even after controlling for MNE presence across industries and countries,[24] confirming the importance of the link between GVCs and GINs.

However, the inclusion of the MNE variable seems to change the results of some variables. In particular, the distance measures and the technological distance become insignificant. However, this may be due to changes in the samples since there is limited availability of MNE data across countries and industries in the OECD AMNE database. The AMNE information is typically more available for EU countries and the United States, for which on average the variability in the above mentioned variables is much smaller[25].

In order to allow for potential temporal effects, the trade variables have been lagged with one to three years as one could argue that familiarity in trading networks is not instantaneous and takes some time before partners getting to know each other better. The results are presented in Annex 2 and confirm the importance of trade relationship for collaboration in innovation across different time periods. However, contrary to what could be expected, the effect of trade on co-invention seems to become somewhat less important over time. One reason for this result could be the changes in trade relationships that happen over time with partners setting up new trade links and stopping existing ones.[26]

Lastly, estimations have been also undertaken using value added exports instead of gross exports: if value added trade would reflect a more intensive relationship between partners than gross trade, a stronger effect co-invention could be expected. Again, the results confirm the link between GVCs and GINs but no significant differences are found in the two types of trade relationships. The results are presented in Annex 3.

4. From GINs to GVCs: does co-invention induce GVC upgrading?

The links between GVCs and GINS may also go in the opposite direction due to the existence international spillovers of technology and knowledge. The endogenous growth literature has discussed the role of international knowledge flows for economic growth: spillovers from investments and activities (in knowledge) in one location and by one party may result in beneficial effects for other economic actors in other locations. Different channels of international spillovers have been analysed, amongst which most prominently trade and foreign direct investment (FDI). Empirical research has shown that imports and exports generate international spillovers as products and services embody technological knowledge (Coe et al., 1997; Keller, 2004). Likewise, the growing trade of intermediates within GVCs reflects the embodiment of the skills, factors and technologies used to produce them and therefore can be a form of technology transfer from abroad to local firms (Amiti and Konings, 2007; Bas and Strauss-Kahn, 2015; Halpern et al., 2015).

International investment may also act as a channel of technology/knowledge diffusion since local firms may learn about advanced technologies or good organisational and managerial practices of foreign MNEs (Caves, 1996). A large empirical literature has investigated FDI spillovers largely showing that positive spillovers emerge particularly through backward linkages; the evidence on spillovers through other MNE linkages is more mixed (see Alfaro (2014) for an overview of the evidence and issues at stake).

A disembodied channel of technology diffusion is the (international) collaboration in innovation between partners. Technological knowledge often includes non-codified and tacit knowledge which is overall more difficult and costly to transfer. Close interactions and face-to-face contacts facilitate the transfer of knowledge and this is largely the reason why spillovers tend to be localised and often require absorptive capacity (Cohen and Levinthal, 1989; Griffith et al., 2004; Lyachagin et al., 2016). Knowledge may also diffuse more rapidly when interpersonal links through joint work and co-operation create opportunities for learning which go beyond the exchange of codified information. Recent research shows that companies and countries that are connected in international collaboration are more innovative (Breschi and Lissoni, 2009; Singh, 2005; Hoekman et al. (2009).

The question addressed in this paper is then whether GINs and co-inventions in patents help countries to innovate and upgrade in GVCs. Will the GVC activities of countries benefit from their interactions in GINs, as collaboration in innovation provides access to foreign knowledge and technology and thus may give rise to international spillovers? The value created in GVCs is unevenly distributed and depends on the ability of participants to supply sophisticated and hard-to-imitate products and services. Increasingly, such products or services stem from forms of KBC such as brands, basic R&D, intellectual property like patents, etc. OECD (2013b) has shown that KBC is an important driver of GVC upgrading, which enables companies and countries to create and capture more value and improve their position in GVCs. This paper aims to extend this discussion and include the effect of international knowledge spillovers (through co-invention patterns) – in addition to own investments and activities in knowledge - on the upgrading of countries within GVCs.

Innovation and KBC are closely linked to each of the different patterns of upgrading – process, product, functional and chain upgrading (OECD, 2013a). In general, upgrading trajectories at the country level become apparent when looking at the domestic value added content of countries' export. This indicator[27] provides insights into how much value an economy creates from its exports and is overall the result of the different upgrading strategies of companies located within its borders. The base empirical model used in this paper links then the change in the domestic content of exports to own investments in KBC (proxied by own patent stock) and international knowledge spillovers (which in line with the empirical spillover literature are calculated as the weighted average of the foreign partners' patent stocks, with the weights being the co-invention patterns across partners[28]):

$$\Delta \frac{dva}{\exp orts_{ist}} = \alpha_0 + \alpha_1 pstock_{ist} + \alpha_2 \sum \left(\frac{coinv_{ijst}}{coinv_{ist}} * pstock_{jst}\right) + \varepsilon \qquad (4)$$

with i = country i, j = partner country j, s = industry and t = year

The results presented in Table 2 primarily demonstrate the importance of own investments in KBC of countries for GVC upgrading; this is in line with OECD (2013a) which showed the positive effect of countries' KBC on the export competitiveness and upgrading of these countries within GVCs. On the contrary, the results do not seem to indicate any effect of GINs via international spillovers on the GVC upgrading of countries. Moreover, after taking into account a number of factors that have been discussed in the spillover literature to matter and mediate international knowledge flows, no real evidence is found for the contribution of co-invention networks on GVC upgrading. First, correcting for the physical and time distance between partners in collaborations (models 1 and 2) – as the previous analysis has shown the importance of both types of distance for co-invention patterns, does not significantly change the results. Second, the types of partners do not appear to play a major role (model 3); international spillovers do not seem to differ between international collaborations with stronger partners and those with more similar partners. Third, taking into account the importance of absorptive capacity (model 4) – through own investments - does neither show a significant effect from GINs to GVCs.[29]

Annex 4 presents the econometric results for the individual industries of chemicals (ISIC 23-26), electronics (ISIC 30-33) and business services (ISIC 70-74), i.e. industries with a significant number of co-inventions. Results for the chemicals and business services industries are in line with the reported results above finding no evidence of an effect of co-invention on the GVC-upgrading in countries. Results for the electronics industry show - surprisingly - rather a negative effect of co-invention on GVC upgrading; more research is needed to identify the reasons behind this, like for example the results that co-invention is more likely between more technologically distant partners

Table 2. Upgrading model: econometric results

	Base model a)	Base model b)	Base model c)	Model 1	Model 2	Model 3	Model 4
Own Knowledge	0.001	0.013***	0.009**	0.009**	0.009**	0.008*	0
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(.)
Coinvention * Partner Knowledge	0.002**	0.001	-0.002				-0.012**
	(0.00)	(0.00)	(0.00)				-(0.01)
Co-invention*Partner Knowledge*Distance				-0.001			
				(0.00)			
Coinvention* Partner Knowledge * Time difference					-0.001		
					(0.00)		
Coinvention * Partner Knowledge * (Partner/Own knowledge)						-0.001	
						(0.00)	
Coinvention * Partner Knowledge*Own Knowledge							0.009**
							(0.00)
Control Variables							
Country*sector		✓	✓	✓	✓	✓	✓
Year			✓	✓	✓	✓	✓
R-squared	0.004	0.43	0.439	0.439	0.439	0.439	0.439
Observations	1,607	1,607	1,607	1,607	1,607	1,607	1,607

Note: level of significance * p<0.10, ** p<0.05, *** p<0.01

Overall the empirical findings in this paper do not support the importance of co-invention within GINs for success in GVCs. However, when interpreting these results a number of important points have to be made, suggesting the need for further research to disentangle the complex interdependencies between GINs and GVCs:

a) the empirical literature on spillovers is broad and extensive and has, to say the least, produced diverse results. Spillovers along the different channels discussed above range from negative in a number of analyses (usually in the case of MNE spillovers in some developing economies) to positive in relatively more studies. A publication bias in spillover results has however been observed with a tendency to publish results if studies are able to reject the null hypothesis, that is, when the investigations produce positive and statistically significant findings (Havranek, 2012; Demena and van Bergeijk, 2016).

b) most of the empirical research on knowledge and technological spillovers has focused on productivity (particularly via trade and FDI) and innovative outcomes (for disembodied channels) while studies on the impact of spillovers on GVC upgrading are non-existent. Indeed, the concept of GVC upgrading is rather broad given the different upgrading patterns; consequently, empirical indicators on GVC upgrading are not uniformly defined[30] and have been used only recently.

c) the spillover literature has shown that timing issues matter in assessing the beneficial effects of international knowledge flows with some spillovers only materialising after quite some time. A general empirical strategy has not been established in order to capture these timing effects, but the availability of TiVA data has somewhat limited the empirical analysis in this paper. Furthermore, the TIVA data encompass relatively different time periods for GVCs, with the periods 1995-2000 and especially 2000-2005 showing a significant expansion of GVCs while the economic crisis has resulted in a consolidation of GVCs in the period 2005-2010.

d) investment in knowledge – own as well as partners' – in this analysis are proxied by patent stocks; as mentioned already above, recent OECD work has shown the need to analyse KBC from a broader perspective beyond R&D and intellectual property. Also the traditional disadvantages of patent data – e.g. differences in patent propensity across industries and countries, have to be taken into account when interpreting the results.

e) the analysis in this paper has been undertaken at the country-industry level which overall may be too broad to capture spillover effects; recent research has shown the importance of firm-level characteristics for assessing the effects of international collaboration in innovation.

5. Conclusions and some policy implications

In exploring the interdependencies between GVCs and GINs at the country level, this paper has reported on a number of clear links between both types of networks. At the same time, as it became clear in the discussion of the results, the analyses in this paper should be regarded as only a first step and more research is needed in order to better and fully understand the linkages between GVCs and GINs. Several directions for future research can be proposed: complementing the aggregate analysis with more firm-level analysis; discussing different types of GINs (the above-mentioned work on the links between GVCs and KBC goes in this direction); and, the discussion of timing issues, etc.

Despite the limitations, the paper has provided evidence on the growing importance of international linkages in the innovation landscape of countries; companies are increasingly opening their innovation process and collaborating with other partners across borders. The number of international co-inventions in patent data shows a gradual increase over the past two decades, although the economic crisis of 2009 has slowed this internationalisation somewhat. Nevertheless, in developing an effective innovation strategy, policy makers will have to increasingly take into account this international perspective; an overly inward-looking perspective in innovation may isolate domestic companies in the global innovation process.

While previous OECD work had already pointed to the growing internationalisation of innovation, this paper additionally shows the importance of MNEs in this development. As almost two-thirds of international co-inventions during the period 1995-2013 are directly linked to MNE activities – and more than 50% of the international co-inventions are held by one MNE, it is clear that GINs are to a large extent structured within and around MNE networks. Also this is an important point to take into account when developing and implementing government policies; connection to such MNE networks by attracting foreign MNEs and helping to grow incumbent companies to become MNEs, needs to be an essential part of an effective government policy.

Another important observation emerging from this paper, is that distance is not dead yet. Even in a time with extensive and advanced communication possibilities, distance still seems to matter. GVCs and GINs both show a strong regional character with trade and co-invention relationships still strongly concentrated within supra-national regionals (North America, Europe and Asia), notwithstanding the growing importance of extra-regional links. Internationalisation strategies and policies will have to take into account both the costs of distance due to face-to-face interaction costs (as reflected by geographic distance) and virtual interaction costs (as reflected by time distance).

Despite the importance of distance, the geography of both GVCs and GINs has been changing with networks becoming denser and linkages stretching across a growing number of countries including emerging economies. While there are clearly some parallels to be drawn between the development of GINs and GVCs in recent years, there are nevertheless a number of clear differences between both types of networks. First, GINs are not as intensive as GVCs; second, GVCs exhibit a more stable pattern as the economic globalisation through trade has already been in place for many decades.

In general, GVCs and GINs show a strong overlap in geographic concentration with hubs in GVCs often also being hubs in GINs. Questions have been raised in policy discussions about the localised impact of STI, i.e. countries are not always benefitting from the advances in STI (including the government support for R&D). The aggregate analysis (on the supra-region and country level) in this paper does not seem to support this claim very strongly. That said, it is clear that a more disaggregated analysis (on the sub-regional and company level) is desirable and necessary to address this important policy issue in more detail.

One link between GVCs and GINs clearly demonstrated in this paper is the strong positive effect reported going from GVCs towards GINs. Partners who are trading with each other – be it via exports or imports - are found to cooperate more intensively also in GINs. Collaborating within a trade relationship offers the possibility to identify new partners and get more familiar with these partners. This directly means that apart from S&T policies, also trade policies can have a role to play for the integration of countries in GINs; more research is needed to assess the relative importance of different government policies.

Indeed, previous OECD work has shown that a broad range of STI policies can promote the (international) co-operation in science, technology and knowledge. As STI policies can no longer be designed solely in a national context, government support for R&D innovation, strong universities and other public research institute, a modern IPR system, etc. all should reflect the increasing open and global character of technology and knowledge. In contrast, the paper was not able to report evidence of the effect of GINs on GVCs; the fact that a country is heavily involved in GINs does not directly help that country to upgrade in GVCs. In the first place, this shows the broader and complex determinants of GVC upgrading in which GINs could play a role. A number of shortcomings have been discussed largely reflecting the difficulty often encountered in empirical work to demonstrate the effects of STI on economic performance and the impacts of international spillovers. As such, the (lack of) evidence in this paper should not be interpreted as an indication of the importance of GINs for the economic performance of companies, industries and countries. Rather, it demonstrates a potential area for future research because its direct importance for policy.

NOTES

1 The first type might be easily assimilated with the material flows of intermediate and final products, while the second one focuses more on the immaterial flow of ideas, knowledge and technology. However, GVCs are broader than supply chains, including also the upstream activities of R&D, design and so on; explaining to some extent the different performance of countries in GVCs.

2 Based on the OECD Inter-Country Input-Output (ICIO) system which links national IO tables of 61 countries across 34 industries.

3 Data on licensing can also be used to document the trend to open innovation but are overall rather limited and lack uniformity. As most patent licensing is based on private contracts that are subject to confidentiality agreements, robust statistics on technology licensing are not available. Furthermore, accounting rules do not require firms to disclose patent licensing revenues as a separate item in corporate reports.

4 Patents protect technological inventions, i.e. products or processes that provide new ways of doing something or new technical solutions to problems.

5 Patent indicators constructed on the basis of information from a single patent office typically suffer from a “home advantage” bias in proportionate to their inventive activity, domestic applicants tend to file more patents in their home country (or region) than non-resident applicants

6 Inventors are physical persons from which it is not clear in which company and industry they work.

7 The reason for using applicants’ information is that inventors are typically physical persons and no direct information is available if they work for or are employees of a MNE; instead, if the applicant is found to be part of a multinational group, the PCT application and co-invention is assumed to belong to MNEs.

8 This has been done by linking information on the ownership structure – available in the ORBIS database –with the HAN database.

9 For about one quarter of the PCT applications as well as international co-inventions, no information on the applicants was included in the HAN and ORBIS; given that both databases are more representative for larger companies including MNEs, one could assume that this category of “no information” concerns relatively more non-MNEs.

10 Some caution has to be taken into account when interpreting these data as the information on the global ultimate owner may not always be available and accurate for some MNEs.

11 Abstraction made from the group of PCT patent that cannot be assigned to one country.

12 Which is of course due to the smaller size of the countries (relative to the United States for example).

13 Co-invention is calculated by adding the sum of co-invention shares between country-pairs divided by the sum of domestic inventions in these respected countries for industry i at time t . Similarly, trade intensity is estimated by adding the exports between respected trading partners out of the sum of output for specific industries at a particular time. While these measures are largely similar, it should be taken into account that co-invention is per se non-directional and largely symmetrical (apart from

the factional counts when more inventors in one country), exports are directional and non-symmetrical.

14 The time periods considered are 2000-2005 and 2008-2011; the choice for the last period is to prevent a possible bias in the data because of the economic crisis in 2008.

15 The co-invention intensity and trade intensity by partners are presented only at the industry level; because of the multitude of bilateral relations, a similar graph on the economy-wide level is not feasible.

16 OECD work has showed that services are to a large extent included in the exports of goods, i.e. the gross value of manufactures exports includes a large share of value added that has been created in services.

17 The gravity model is the workhorse model of international trade, going back to a contribution by Tinbergen (1962); see Anderson and van Wincoop (2003) for a discussion of the theoretical foundations of gravity modelling.

18 Although OLS estimations are less appropriate given the large number of zeros in the dependent variable, the empirical model has also been estimated using OLS as robustness test. The OLS results (not reported here) are largely in line with the PPML results and subscribe in particular the importance of trade relations between countries for co-invention.

19 Using patent stocks as indicator proxy for countries' innovation activities represents only a (rough) proxy as recent OECD work on Knowledge Based Capital (KBC) has shown that innovation is much broader than intellectual property (OECD, 2013b).

20 Another problem is directly linked to the disadvantages of patent data as patenting is not important in all industries; R&D could be used as an alternative indicator of KBC but also R&D data are less available for emerging economies.

21 Source: CEPII database based on km between capitals.

22 As the variables are expressed in logarithms, a dummy variable is included to retain observations between country pairs in the same time zone.

23 The different relationship appear robust to the inclusion of both year and importer (partner-industry) and exporter (country-industry) fixed effects

24 Because of more limited data availability of MNE presence, the analysis takes necessarily into account a smaller number of observations.

25 Also the no-trade variable is dropped from the model as all observations with positive MNE turnover show also positive bilateral trade.

26 Nevertheless, previous research has shown that changes in trade happen largely at the intensive margin (changes in exports/imports volumes with existing partners) rather than at the extensive margin (new partners).

27 It should however be stressed that this is only one possible indicator of GVC upgrading; there is a broad discussion how upgrading can be assessed by which indicators and at what level of analysis (country, industry, company, etc.).

[28] More specifically, the weights – calculated at the industry level - are the shares of the co-inventions of the country in question with each partners divided by all the co-inventions of that country across partners. Because of the availability of the TiVA data, variables have been aggregated over 5 years (1995-2000, 2000-2005 and 2005-2010).

[29] Taking into account the individual and interaction effects simultaneously.

[30] Additional estimations were undertaken with the growth in domestic value added in exports (in absolute terms) and the growth in value added (irrespective if the value added was created in exporting activities or activities for the domestic market); the results of international knowledge flows for GVC upgrading were however also for these dependent variables, non-significant.

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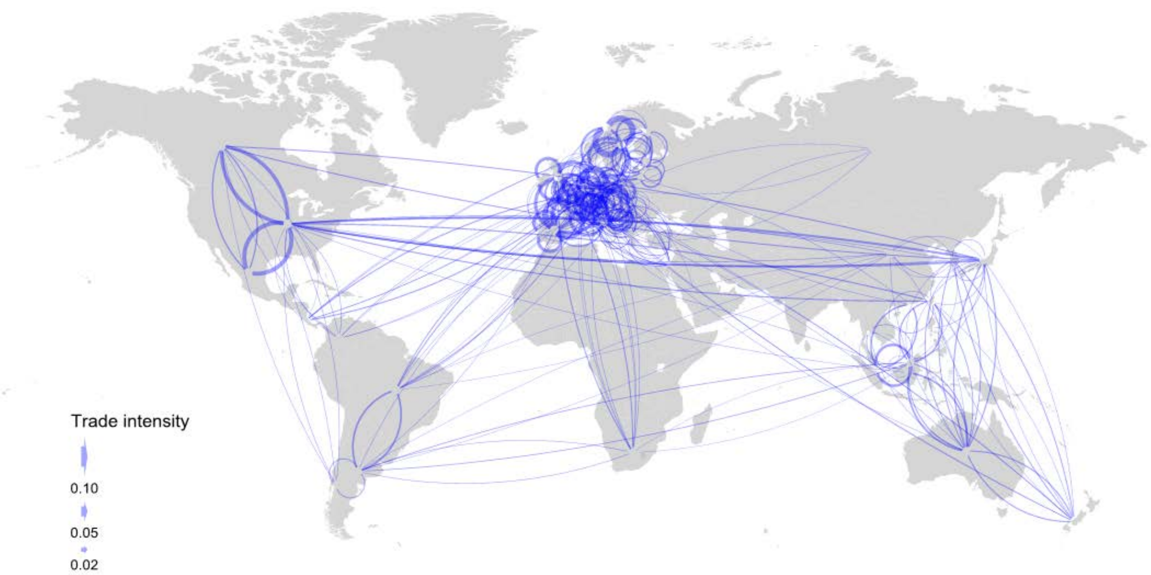
ANNEX 1: CO-INVENTION AND TRADE INTENSITY, BY COUNTRY, PARTNER AND INDUSTRY

Agriculture, hunting, forestry and fishing (ISIC 01 to 05), 2000-05

International co-invention intensity[1]

Trade intensity[2]



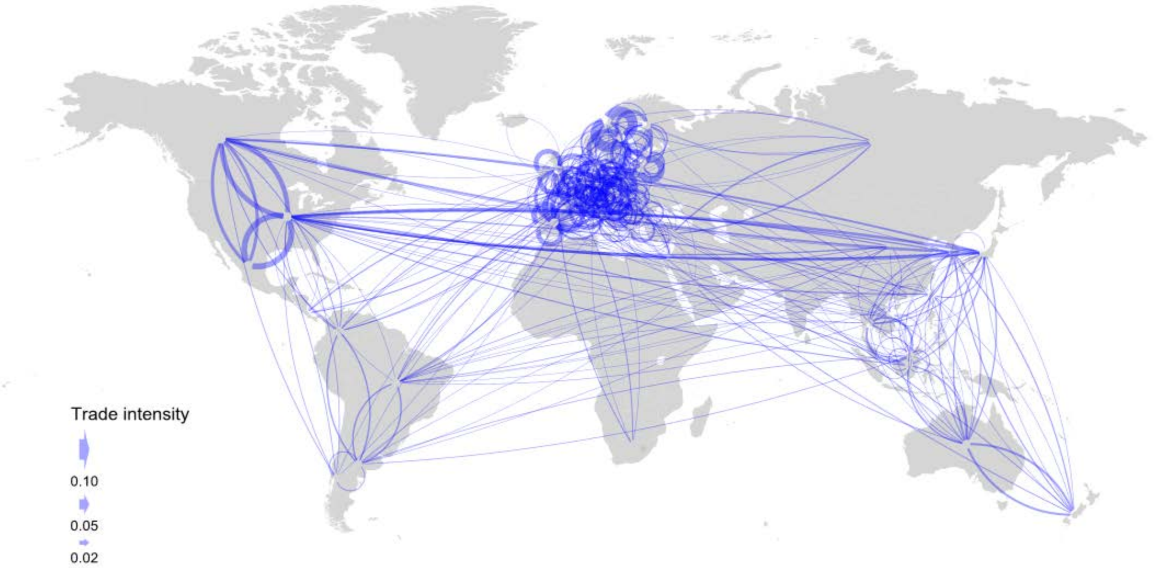
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Agriculture, hunting, forestry and fishing (ISIC 01 to 05), 2008-11

International co-invention intensity[1]

Trade intensity[2]



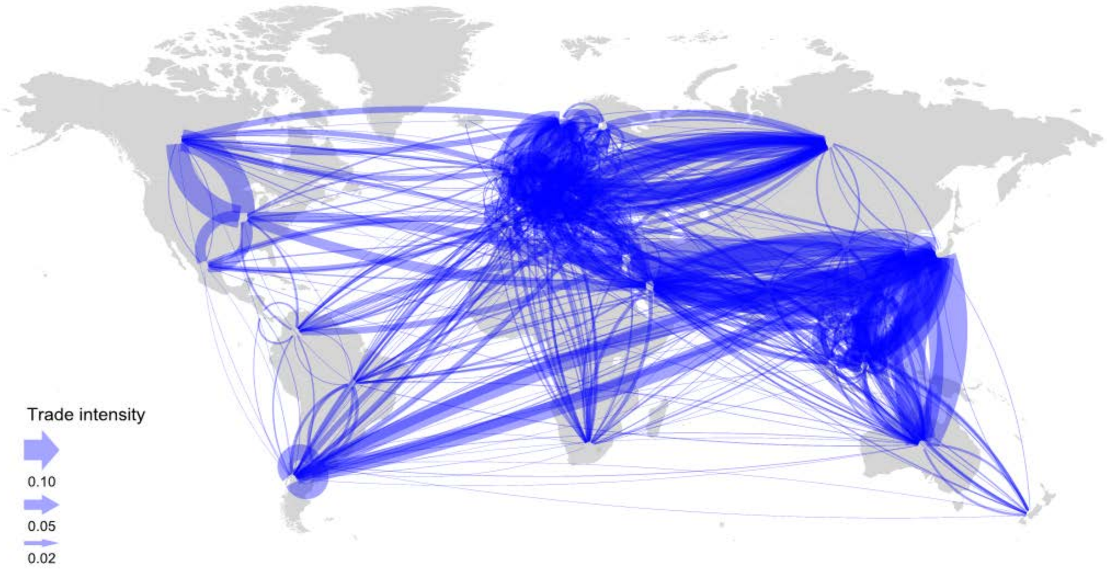
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Mining and quarrying (ISIC 10 to 14), 2000-05

International co-invention intensity[1]

Trade intensity[2]



1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Mining and quarrying (ISIC 10 to 14), 2008-11

International co-invention intensity[1]

Trade intensity[2]

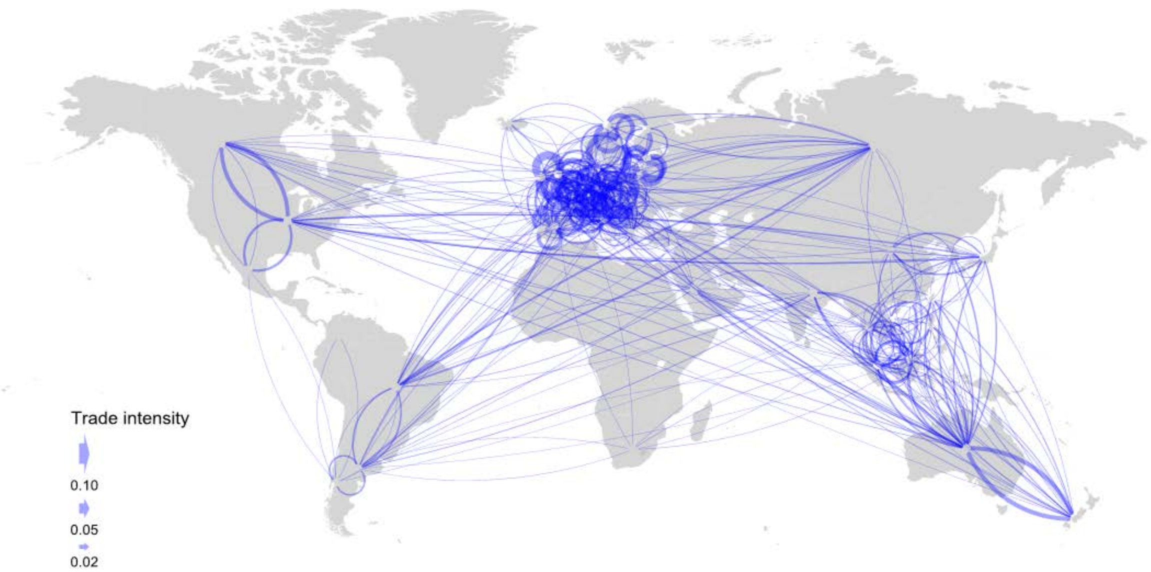
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Food products, beverages and tobacco (ISIC 15 to 16), 2000-05

International co-invention intensity[1]

Trade intensity[2]



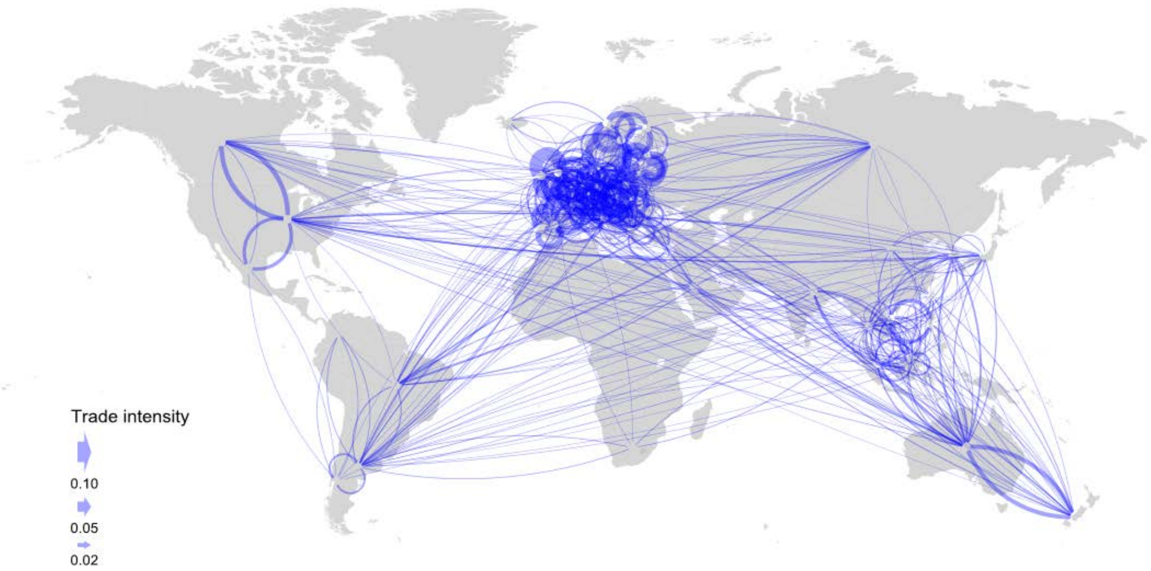
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Food products, beverages and tobacco (ISIC 15 to 16), 2008-11

International co-invention intensity[1]

Trade intensity[2]

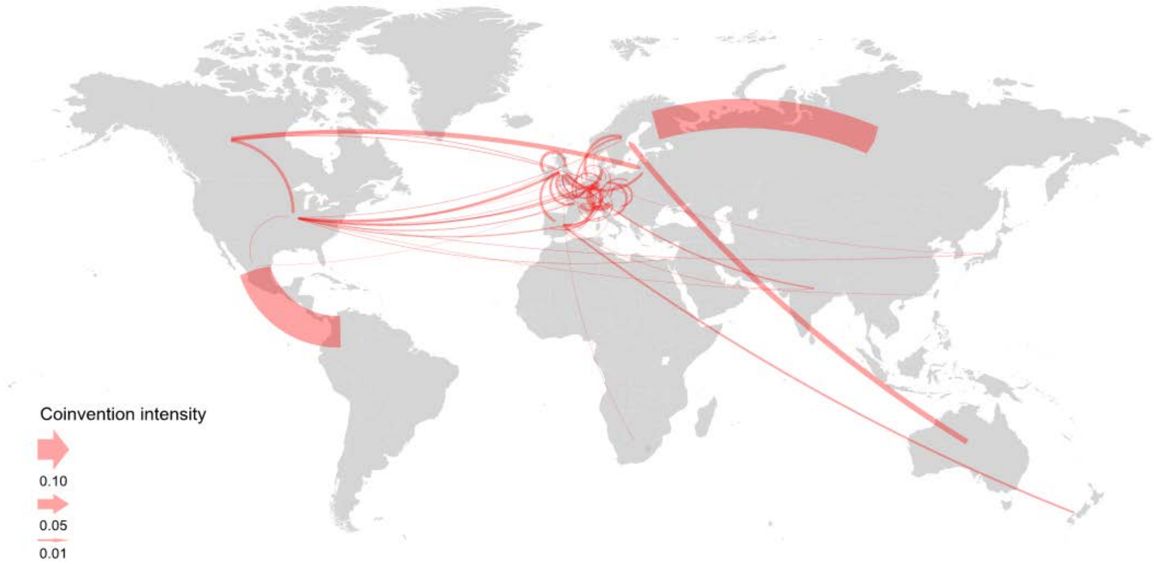


1. Bilateral co-inventions as a share of all inventions by both countries.

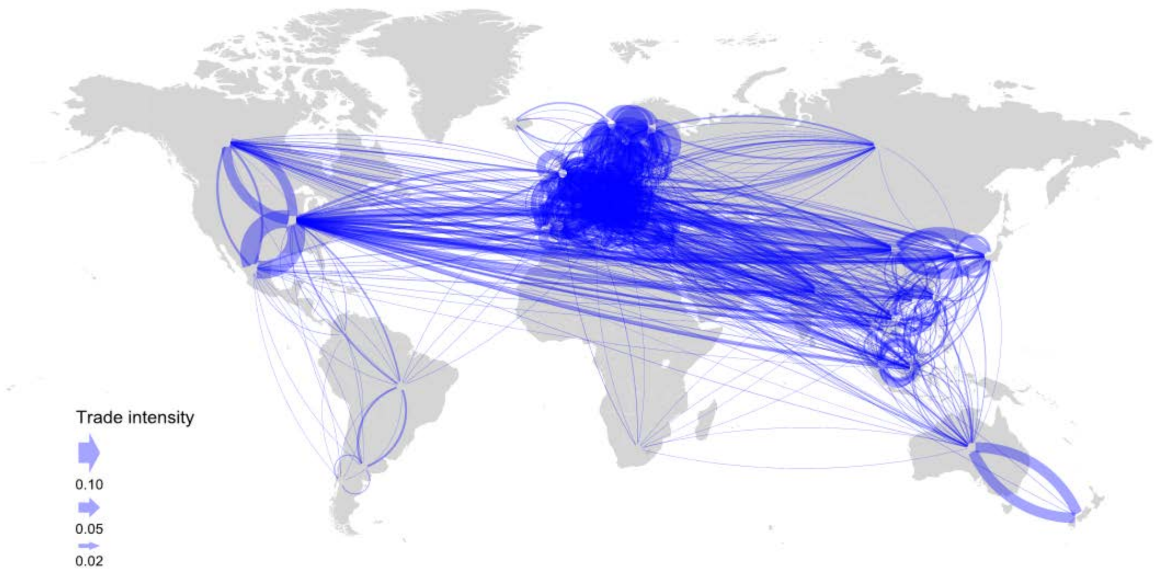
2. Bilateral trade as a share of industry output of both countries.

Textiles, textile products, leather and footwear (ISIC 17 to 19), 2000-05

International co-invention intensity[1]



Trade intensity[2]

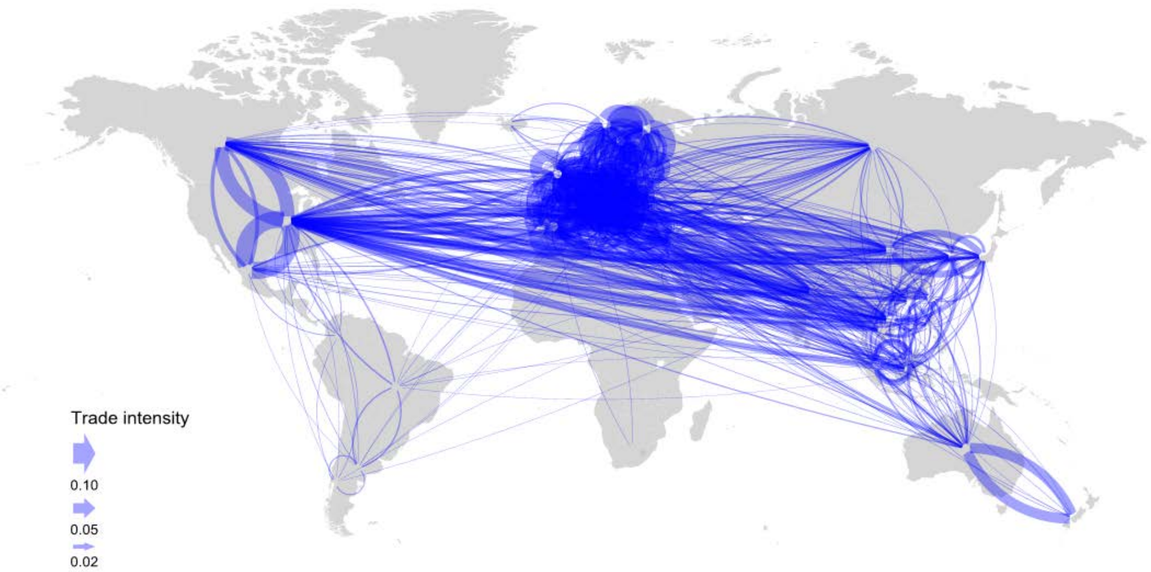


1. Bilateral co-inventions as a share of all inventions by both countries.
2. Bilateral trade as a share of industry output of both countries.

Textiles, textile products, leather and footwear (ISIC 17 to 19), 2008-11

International co-invention intensity[1]

Trade intensity[2]



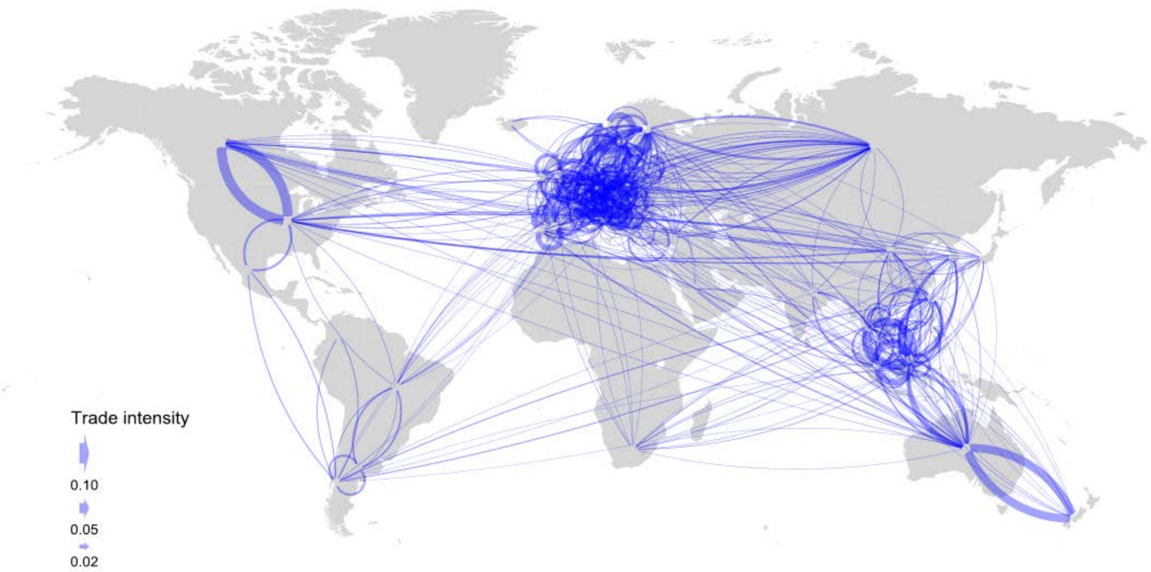
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Wood, paper, paper products, printing and publishing (ISIC 20 to 22), 2000-05

International co-invention intensity[1]

Trade intensity[2]



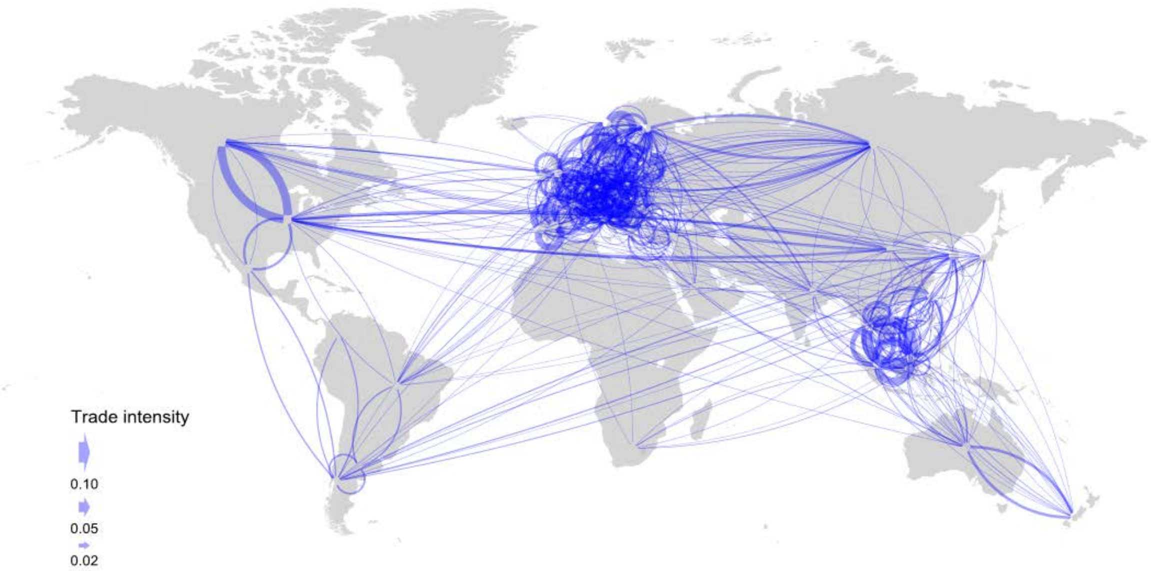
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Wood, paper, paper products, printing and publishing (ISIC 20 to 22), 2008-11

International co-invention intensity[1]

Trade intensity[2]



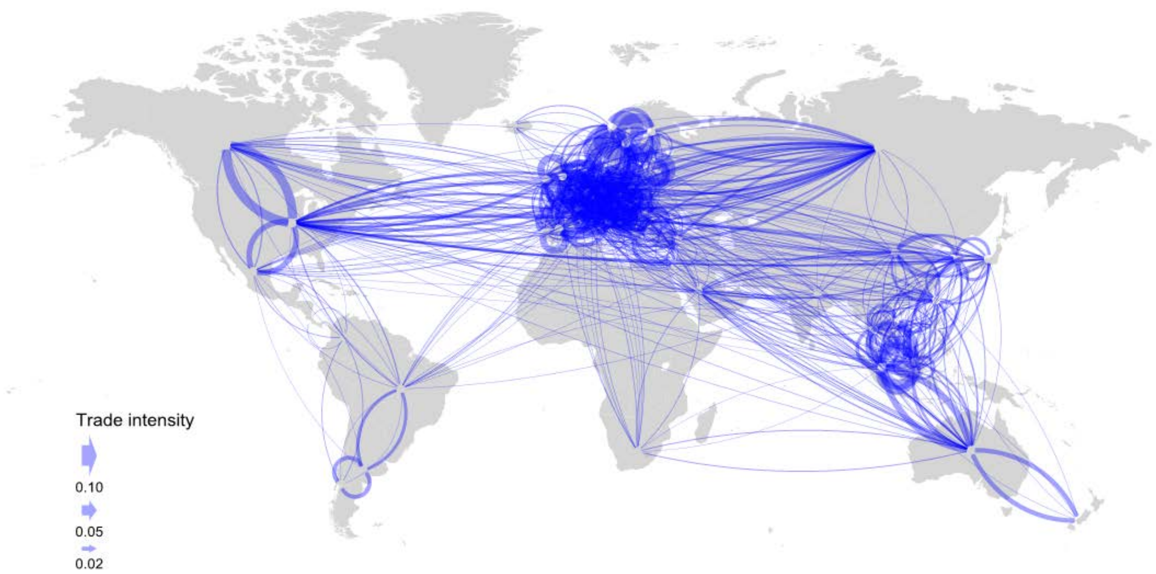
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Chemicals and non-metallic mineral products (ISIC 23 to 26), 2000-05

International co-invention intensity[1]

Trade intensity[2]



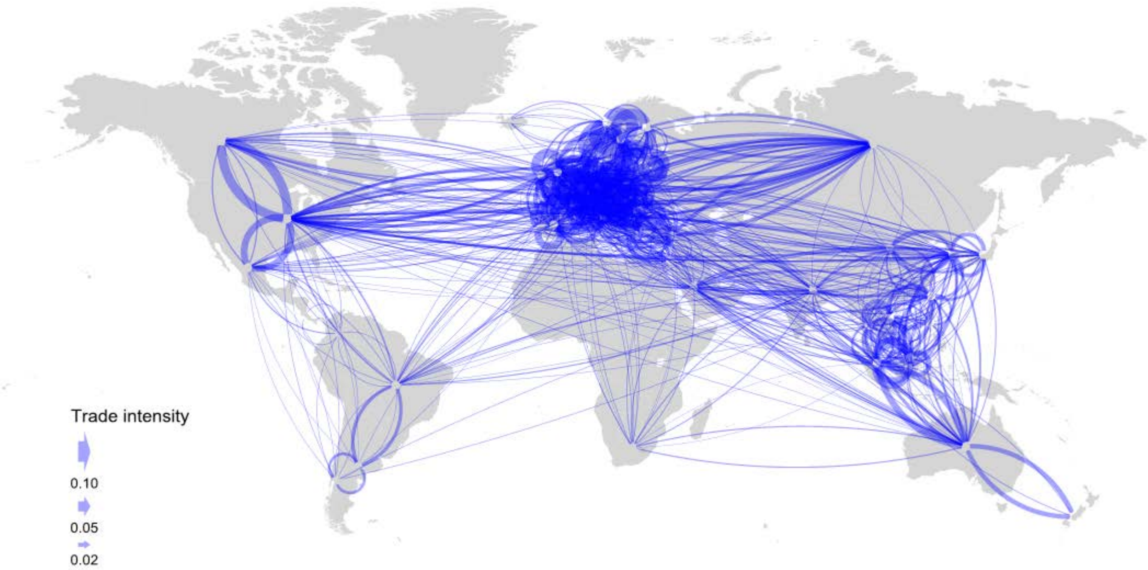
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Chemicals and non-metallic mineral products (ISIC 23 to 26), 2008-11

International co-invention intensity[1]

Trade intensity[2]



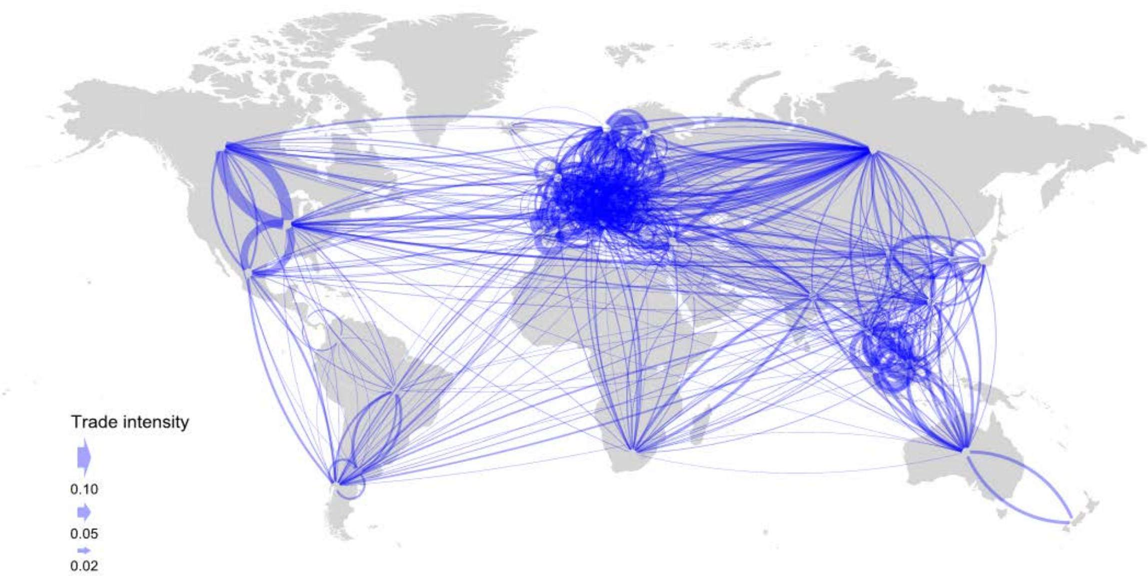
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Basic metals and fabricated metal products (ISC 27 to 28), 2000-05

International co-invention intensity[1]

Trade intensity[2]



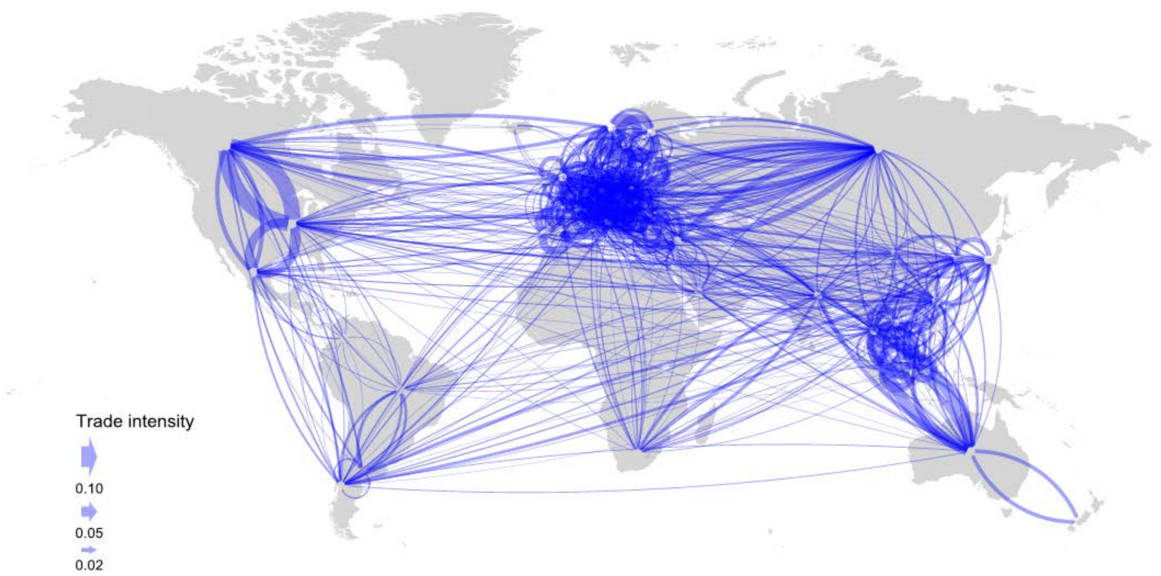
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Basic metals and fabricated metal products (ISC 27 to 28), 2008-11

International co-invention intensity[1]

Trade intensity[2]



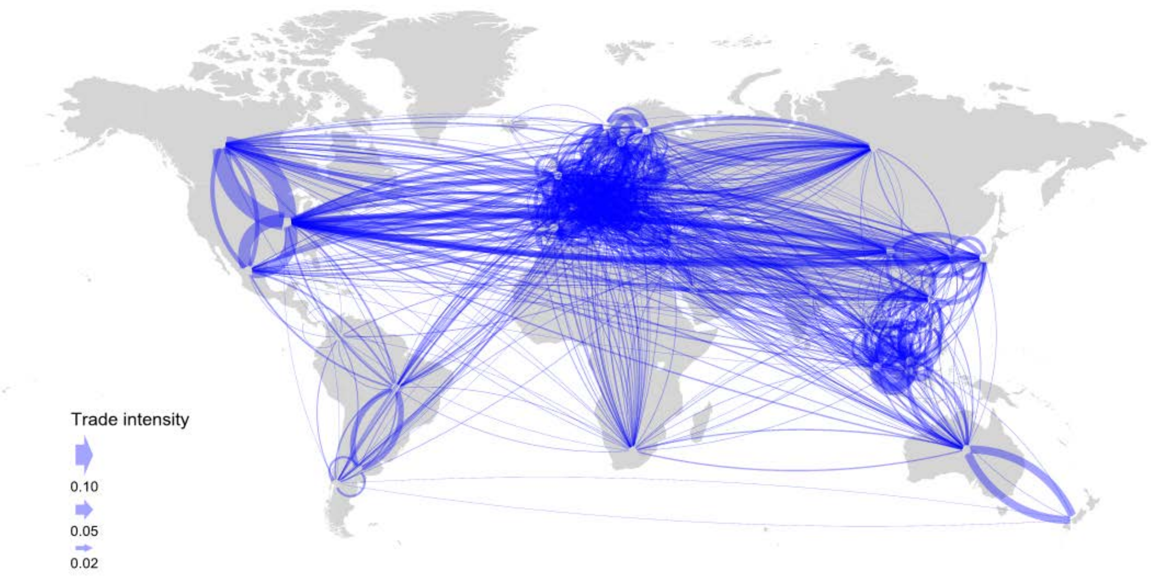
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Machinery and equipment, nec (ISIC 29), 2000-05

International co-invention intensity[1]

Trade intensity[2]



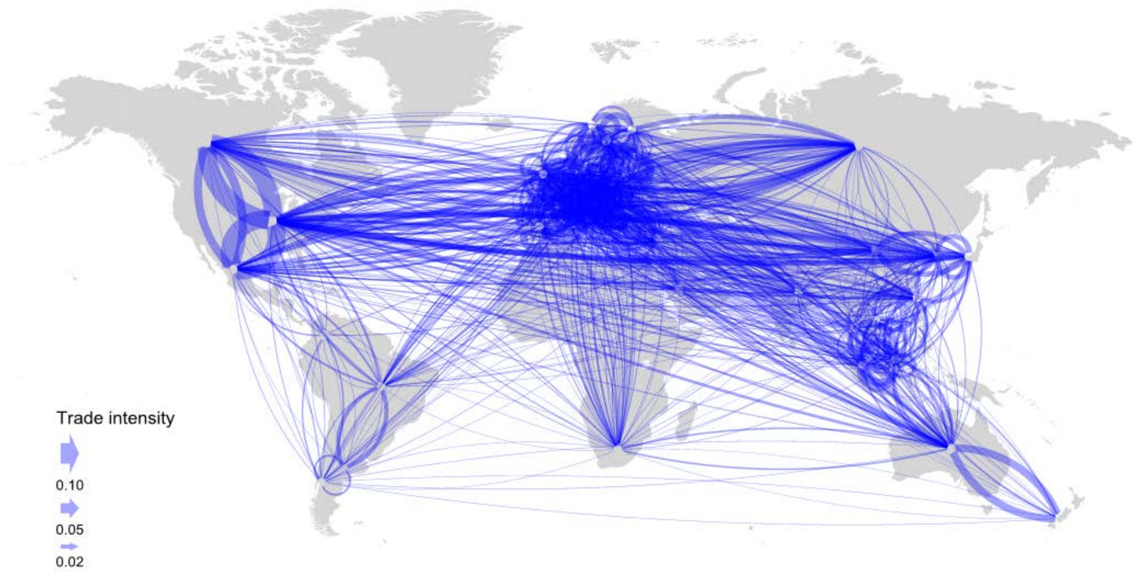
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Machinery and equipment, nec (ISIC 29), 2008-11

International co-invention intensity[1]

Trade intensity[2]



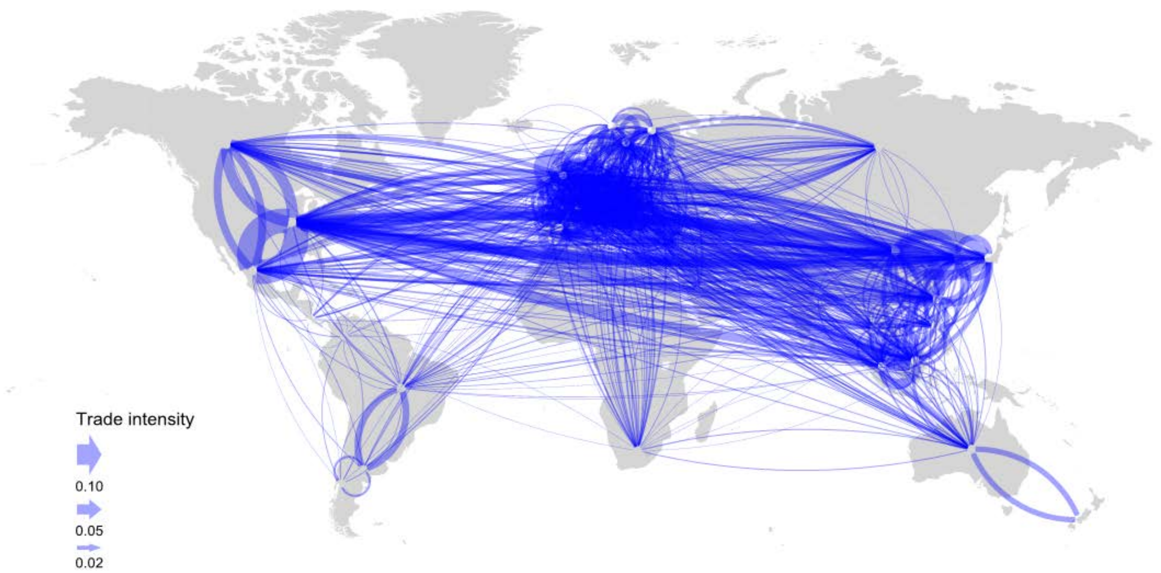
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Electrical and optical equipment (ISIC 30 to 33), 2000-05

International co-invention intensity[1]

Trade intensity[2]



1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Electrical and optical equipment (ISIC 30 to 33), 2008-11

International co-invention intensity[1]

Trade intensity[2]

1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Transport equipment (ISIC 34 to 35), 2000-05

International co-invention intensity[1]

Trade intensity[2]

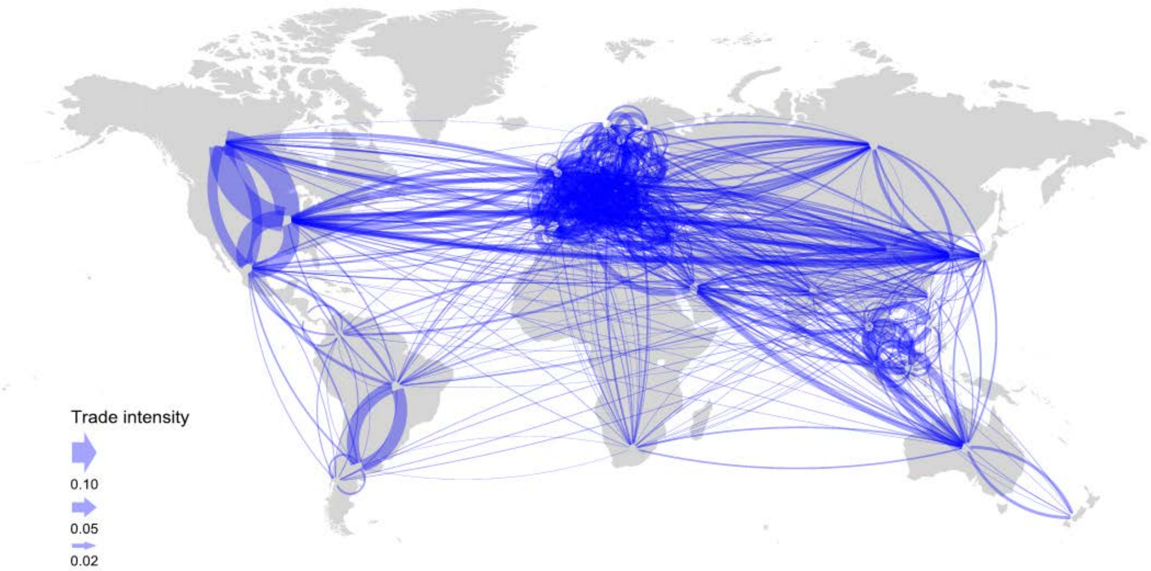
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Transport equipment (ISIC 34 to 35), 2008-11

International co-invention intensity[1]

Trade intensity[2]



1. Bilateral co-inventions as a share of all inventions by both countries.
2. Bilateral trade as a share of industry output of both countries.

Manufacturing nec; recycling (ISIC 36 to 37), 2000-05

International co-invention intensity[1]

Trade intensity[2]

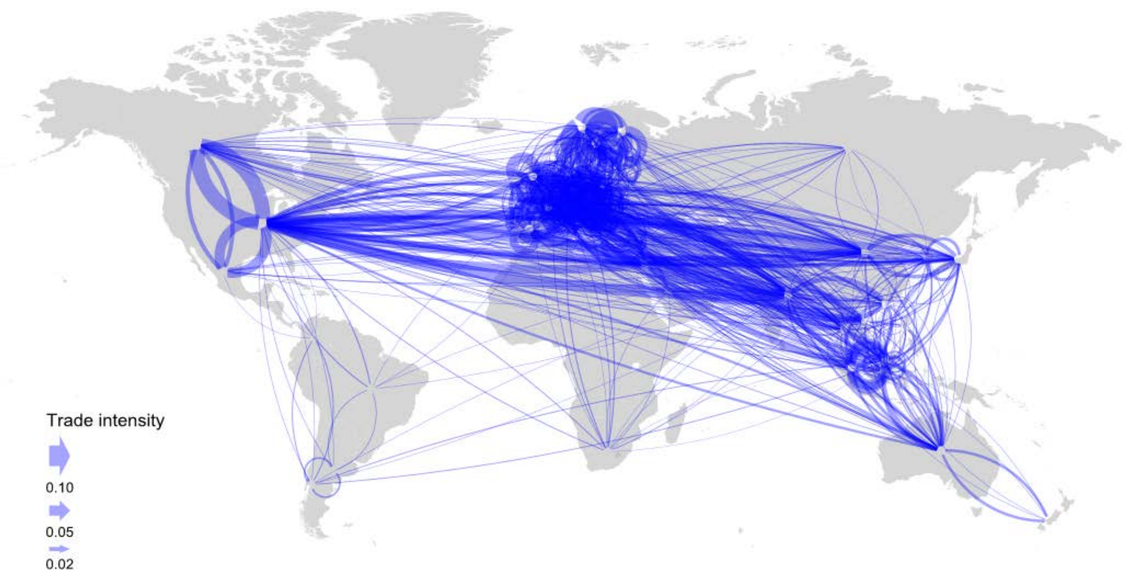
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Manufacturing nec; recycling (ISIC 36 to 37), 2008-11

International co-invention intensity[1]

Trade intensity[2]



1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Electricity, gas and water supply (ISIC 40 to 41), 2000-05

International co-invention intensity[1]

Trade intensity[2]

1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Electricity, gas and water supply (ISIC 40 to 41), 2008-11

International co-invention intensity[1]

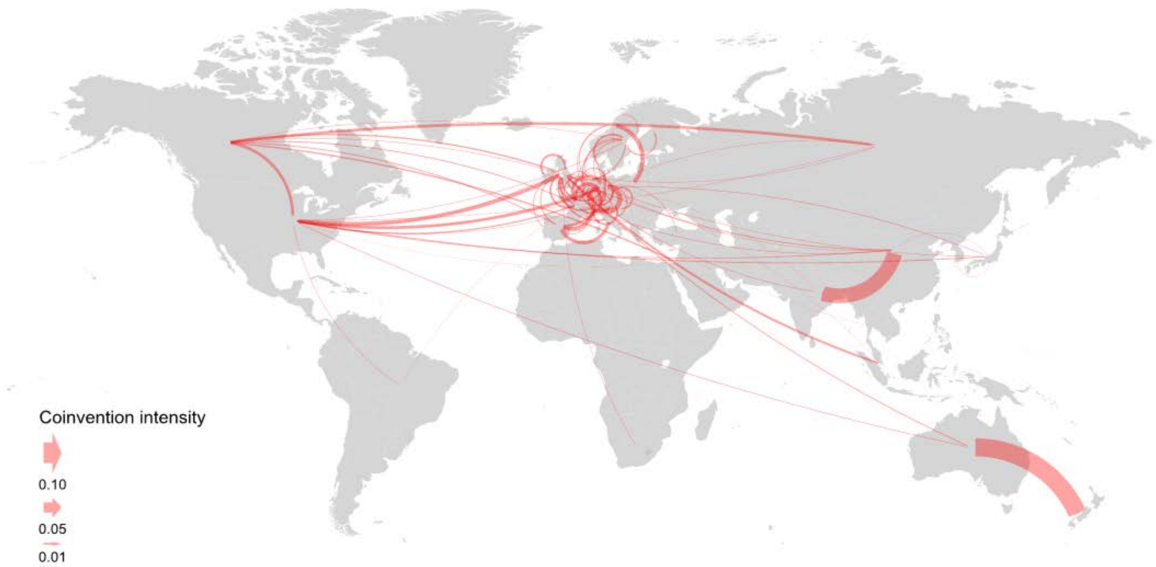
Trade intensity[2]

1. Bilateral co-inventions as a share of all inventions by both countries.

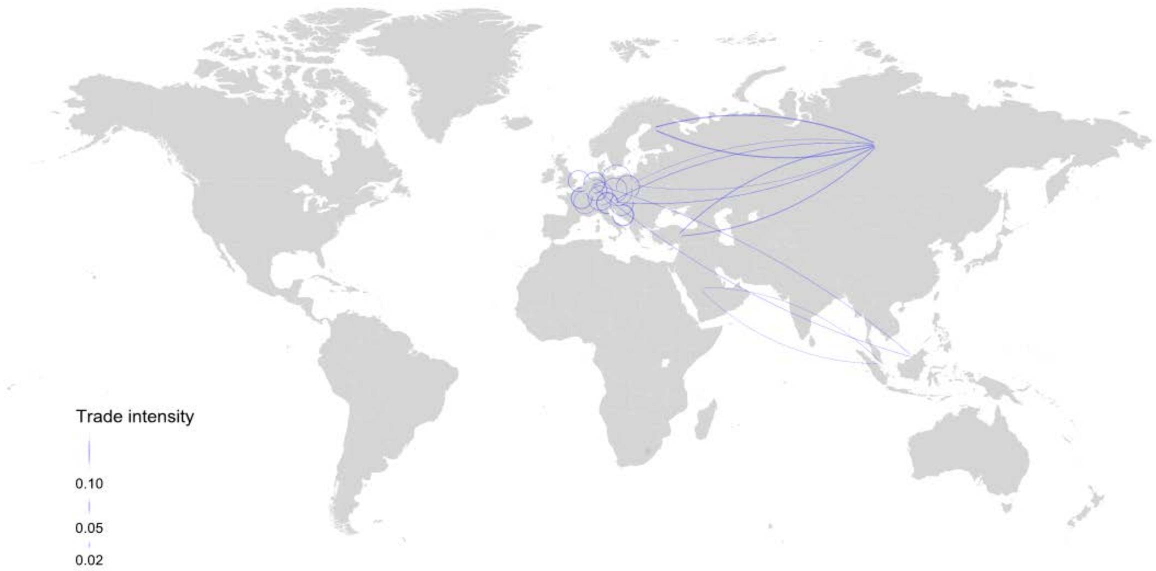
2. Bilateral trade as a share of industry output of both countries.

Construction (ISIC 45), 2000-05

International co-invention intensity[1]



Trade intensity[2]



1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Construction (ISIC 45), 2008-11

International co-invention intensity[1]

Trade intensity[2]

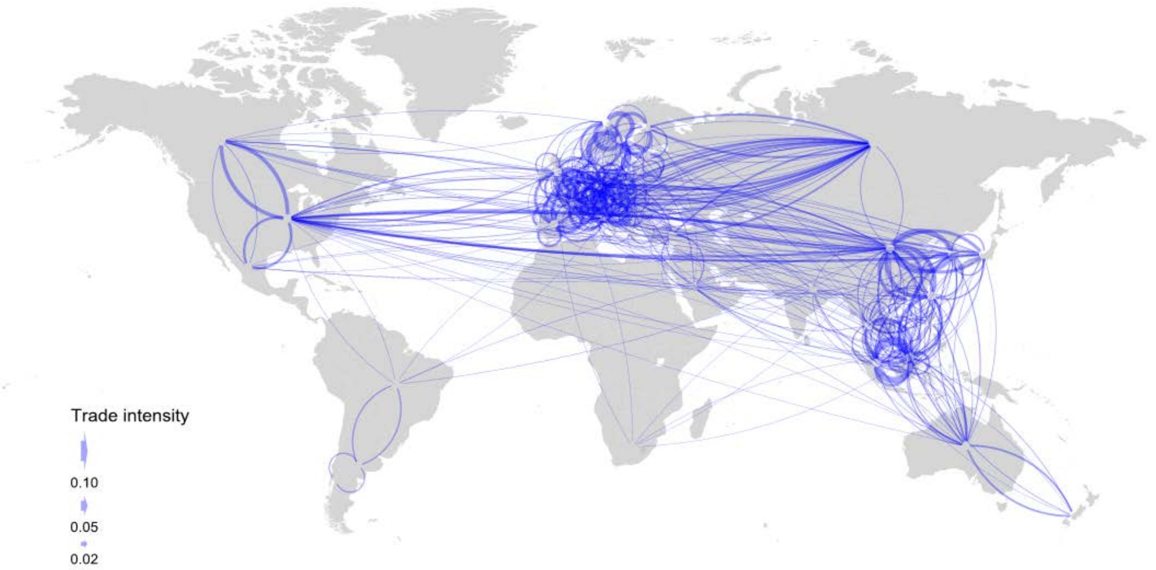
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Wholesale and retail trade; Hotels and restaurants (ISIC 50 to 55), 2000-05

International co-invention intensity[1]

Trade intensity[2]



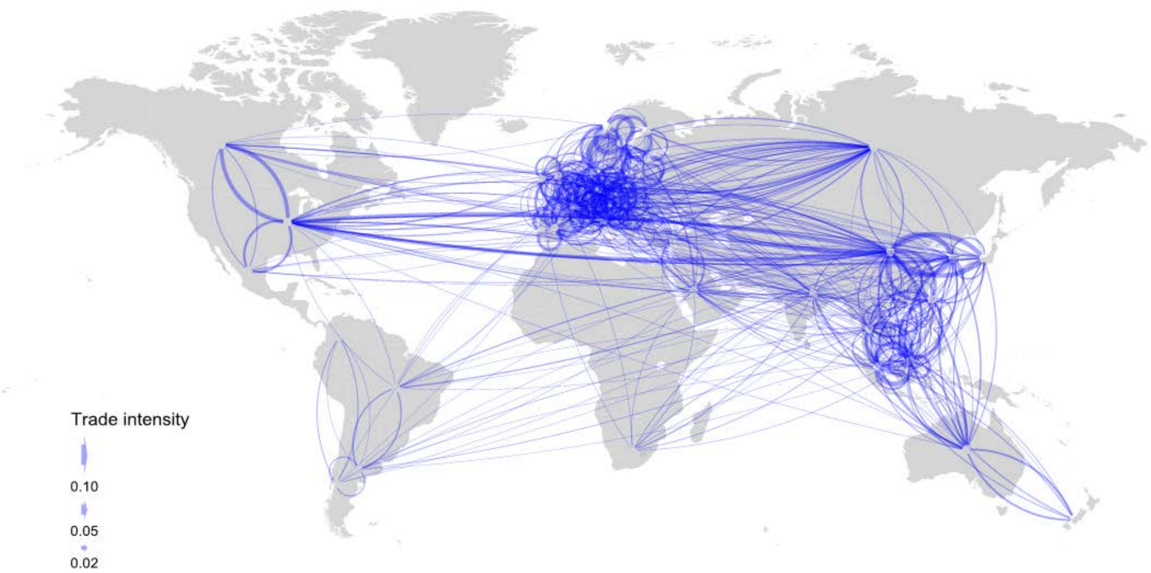
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Wholesale and retail trade; Hotels and restaurants (ISIC 50 to 55), 2008-11

International co-invention intensity[1]

Trade intensity[2]

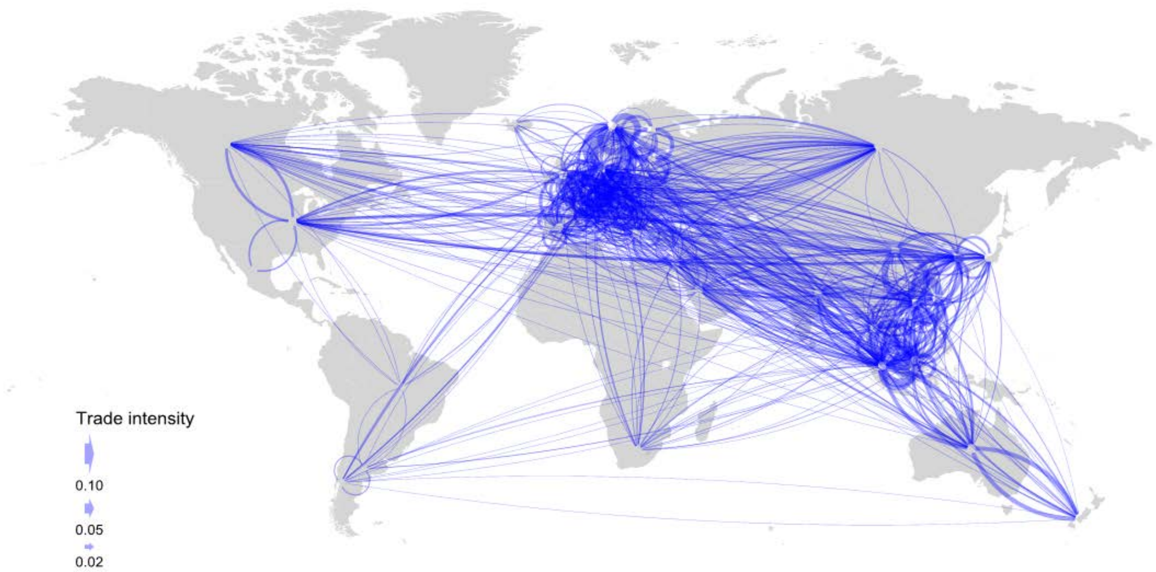


1. Bilateral co-inventions as a share of all inventions by both countries.
2. Bilateral trade as a share of industry output of both countries.

Transport and storage, post and telecommunication (ISIC 60 to 64), 2000-05

International co-invention intensity[1]

Trade intensity[2]



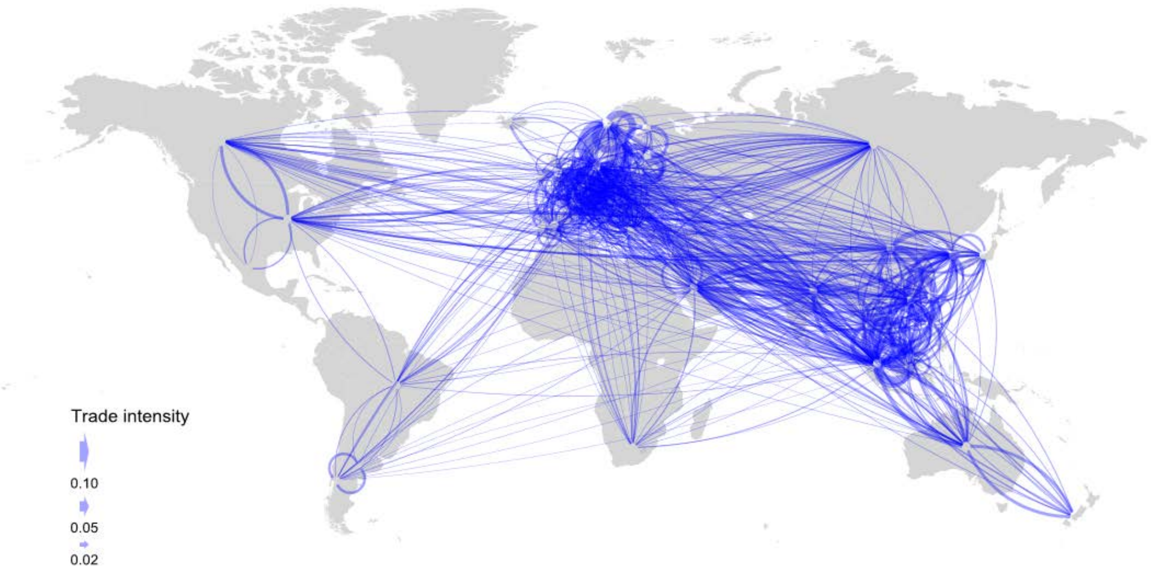
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Transport and storage, post and telecommunication (ISIC 60 to 64), 2008-11

International co-invention intensity[1]

Trade intensity[2]



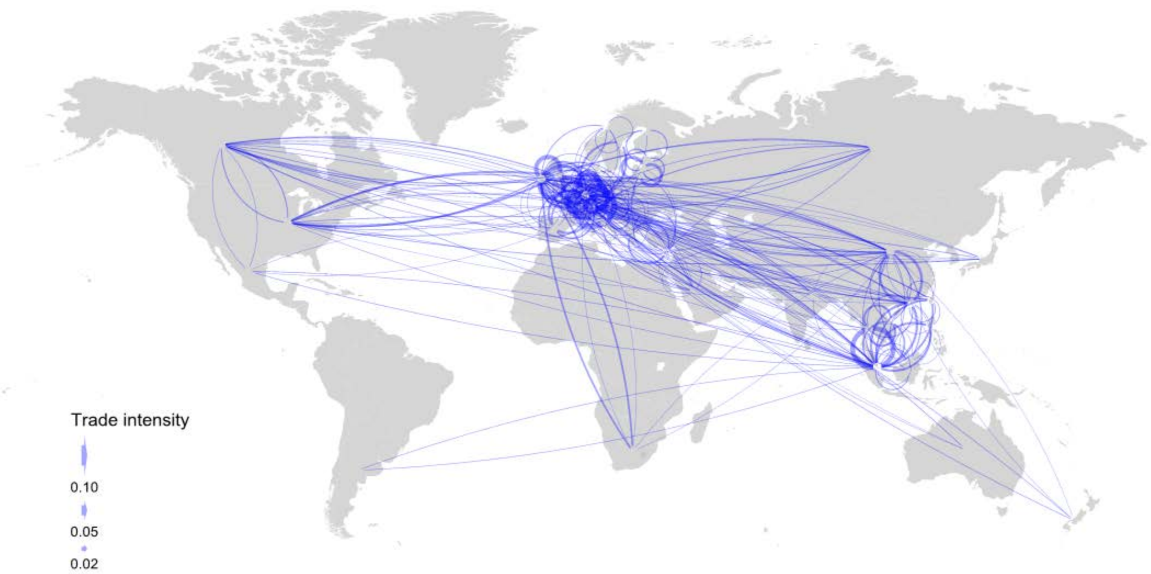
1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Financial intermediation (ISIC 65 to 67), 2000-05

International co-invention intensity[1]

Trade intensity[2]



1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Financial intermediation (ISIC 65 to 67), 2008-11

International co-invention intensity[1]

Trade intensity[2]

1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Real estate, renting and business activities (ISIC 70 to 74), 2000-05

International co-invention intensity[1]

Trade intensity[2]

1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Real estate, renting and business activities (ISIC 70 to 74), 2008-11

International co-invention intensity[1]

Trade intensity[2]

1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Community, social and personal services (ISIC 75 to 95), 2000-05

International co-invention intensity[1]

Trade intensity[2]

1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

Community, social and personal services (ISIC 75 to 95), 2008-11

International co-invention intensity[1]

Trade intensity[2]

1. Bilateral co-inventions as a share of all inventions by both countries.

2. Bilateral trade as a share of industry output of both countries.

ANNEX 2: CO-INVENTION GRAVITY MODEL WITH DIFFERENT TIMING OF TRADE

Timing of trade	t=1	t-1	t-2	t-3
Bilateral Trade	0.269***	0.203***	0.126***	0.071***
	(0.01)	(0.01)	(0.01)	(0.01)
No trade	-3.847***	-3.936***	-2.428***	-0.642
	(0.75)	(0.74)	(0.52)	(0.41)
Patent Stock country	0.469***	0.453***	0.465***	0.492***
	(0.03)	(0.03)	(0.03)	(0.03)
Patent Stock partner	0.271***	0.257***	0.280***	0.308***
	(0.04)	(0.04)	(0.03)	(0.03)
Physical distance	-0.035*	-0.051***	-0.071***	-0.087***
	(0.02)	(0.02)	(0.02)	(0.02)
Time difference	-0.041**	-0.070***	-0.102***	-0.125***
	(0.02)	(0.02)	(0.02)	(0.02)
Same time zone	0.159***	0.155***	0.151***	0.149***
	(0.03)	(0.03)	(0.03)	(0.03)
Technological distance	0.015***	0.013**	0.011*	0.008
	(0.01)	(0.01)	(0.01)	(0.01)
Contiguity (dummy)	0.370***	0.392***	0.416***	0.434***
	(0.03)	(0.03)	(0.03)	(0.03)
Language (dummy)	0.665***	0.689***	0.716***	0.736***
	(0.03)	(0.03)	(0.03)	(0.03)
Colony (dummy)	-0.118***	-0.116***	-0.114***	-0.115***
	(0.03)	(0.03)	(0.03)	(0.03)
Control Variables				
Year	✓	✓	✓	✓
Origin/Destination fe's	✓	✓	✓	✓
R-squared	0.891	0.889	0.888	0.888
Observations	173,918	173,918	173,918	173,918

Note: No trade is a dummy which equals to one when there is zero trade from a country to a partner in a particular industry and zero otherwise. Significance: * p<0.10, ** p<0.05, *** p<0.01

ANNEX 3: CO-INVENTION GRAVITY MODEL WITH VALUE ADDED TRADE

	Base model a)	Base model b)	Base model c)	Model 1	Model 2	Model 3	Model 4
Patent Stock country	0.500***	0.527***	0.468***	0.468***	0.468***	0.464***	0.477***
	(0.01)	(0.01)	(0.03)	(0.03)	(0.03)	(0.03)	(0.06)
Patent Stock partner	0.443***	0.468***	0.269***	0.269***	0.269***	0.266***	0.367***
	(0.01)	(0.01)	(0.04)	(0.04)	(0.04)	(0.04)	(0.06)
Distance	-0.110***	-0.110***	-0.077***		-0.034*	-0.035*	-0.039
	(0.01)	(0.01)	(0.01)		(0.02)	(0.02)	(0.03)
Time difference				-0.064***	-0.039**	-0.039**	0.037
				(0.01)	(0.02)	(0.02)	(0.04)
Same time difference				0.164***	0.161***	0.162***	0.104**
				(0.03)	(0.03)	(0.03)	(0.05)
Contiguity (dummy)	0.314***	0.318***	0.384***	0.408***	0.380***	0.372***	0.370***
	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.05)
Language (dummy)	0.934***	0.953***	0.704***	0.666***	0.667***	0.662***	0.599***
	(0.03)	(0.03)	(0.02)	(0.03)	(0.03)	(0.03)	(0.04)
Colony (dummy)	0.042	0.014	-0.104***	-0.123***	-0.117***	-0.118***	-0.033
	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)
Technological distance						0.015**	0.001
						(0.01)	(0.01)
MNE Turnover							0.014*
							(0.01)
DVA in Trade	0.300***	0.280***	0.278***	0.274***	0.270***	0.273***	0.339***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
No Dva in Trade	-3.053***	-2.985***	-3.988***	-3.951***	-3.957***	-3.954***	
	(0.71)	(0.71)	(0.75)	(0.74)	(0.75)	(0.75)	
Control Variables							
Year		✓	✓	✓	✓	✓	✓
Origin/Destination fe's			✓	✓	✓	✓	✓
R-squared	0.652	0.664	0.889	0.891	0.891	0.891	0.926
Observations	199,534	199,534	174,132	174,132	174,132	173,918	35,665

Note: No DVA Trade is a dummy which equals to one when there is zero DVA in trade between country and partner pairs in a particular industry and zero otherwise. Significance: * p<0.10, ** p<0.05, *** p<0.01

ANNEX 4: UPGRADING MODEL FOR INDIVIDUAL INDUSTRIES

Chemicals

	Base model a)	Base model b)	Base model c)	Model 1	Model 2	Model 3	Model 4
Own Knowledge	0.000	0.025*	-0.015	-0.015	-0.014	-0.019	0
	(0.00)	(0.01)	(0.02)	(0.02)	(0.02)	(0.03)	(.)
Coinvention*Partner Knowledge	0.015**	0.017	-0.001				0.014
	(0.01)	(0.01)	(0.01)				(0.02)
Coinvention*Partner Knowledge *Distance				0.000			
				(0.01)			
Coinvention*Partner Knowledge *Time					0.001		
					(0.01)		
Coinvention*Partner Knowledge *(Partner/Own knowledge)						-0.002	
						(0.01)	
Coinvention*Partner Knowledge *Own knowledge							-0.015
							(0.02)
Control Variables							
Country		✓	✓	✓	✓	✓	✓
Year			✓	✓	✓	✓	✓
R-squared	0.031	0.327	0.371	0.371	0.371	0.372	0.371
Observations	135	135	135	135	135	135	135

Note: level of significance * p<0.10, ** p<0.05, *** p<0.01

Electronics

	Base model a)	Base model b)	Base model c)	Model 1	Model 2	Model 3	Model 4
Own Knowledge	0.011**	0.080***	0.065***	0.068***	0.066**	0.041	0.122***
	(0.00)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)	(0.03)
Coinvention*Partner Knowledge	0.019	-0.039*	-0.058**				0
	(0.01)	(0.02)	(0.03)				(.)
Coinvention*Partner Knowledge *Distance				-0.008			
				(0.03)			
Coinvention*Partner Knowledge *Time					-0.018		
					(0.03)		
Coinvention*Partner Knowledge *(Partner/Own knowledge)						-0.027*	
						(0.01)	
Coinvention*Partner Knowledge *Own knowledge							-0.058**
							(0.03)
Control Variables							
Country		✓	✓	✓	✓	✓	✓
Year			✓	✓	✓	✓	✓
R-squared	0.076	0.472	0.481	0.45	0.453	0.474	0.481
Observations	124	124	124	124	124	124	124

Note: level of significance * p<0.10, ** p<0.05, *** p<0.01

Business services

	Base model a)	Base model b)	Base model c)	Model 1	Model 2	Model 3	Model 4
Own Knowledge	0.001	0.014*	0.010	0.011	0.010	0.004	0
	(0.00)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(.)
Coinvention*Partner Knowledge	0.000	-0.008	-0.012				-0.022
	(0.00)	(0.01)	(0.01)				-0.01
Coinvention*Partner Knowledge *Distance				0.001			
				(0.01)			
Coinvention*Partner Knowledge *Time					-0.002		
					(0.01)		
Coinvention*Partner Knowledge *(Partner/Own knowledge)						-0.006	
						(0.00)	
Coinvention*Partner Knowledge *Own knowledge							0.010
							(0.01)
Control Variables							
Country		✓	✓	✓	✓	✓	✓
Year			✓	✓	✓	✓	✓
R-squared	0.002	0.401	0.404	0.39	0.391	0.402	0.404
Observations	152	152	152	152	152	152	152
Note: level of significance * p<0.10, ** p<0.05, *** p<0.01							